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Reinforced porcelain panel product fabrication methods for enhanced structural protection

Abstract

The present disclosure provides embodiments of a method a reinforced porcelain panel product for renovations or new constructions. In an embodiment, a method includes applying an adhesive to a top surface of a structural core board that includes a foam-core material. Further, the method continues with abuttingly contacting the structural core board to a bottom surface of a porcelain slab with the adhesive such that regions of the structural core board substantially extend beyond outer peripheries of the porcelain slab to define buffer regions, thereby to allow the buffer regions to be visible and to enhance protection from cracking when the reinforced porcelain panel product is handled. Additional features include the bottom surface of the structural core board having a second surface area larger than the first surface area of the porcelain slab and a fracture toughness greater than the porcelain slab.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional application Ser. No. 19/013,292, filed Jan. 8, 2025, titled “REINFORCED PORCELAIN PANEL PRODUCT FABRICATION METHODS FOR ENHANCED STRUCTURAL PROTECTION,” which is a divisional of U.S. Non-Provisional application Ser. No. 18/902,121, filed Sep. 30, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT FABRICATION METHODS FOR ENHANCED STRUCTURAL PROTECTION,” now U.S. Pat. No. 12,241,258, issued Mar. 4, 2025, which is a continuation-in-part of U.S. Non-Provisional application Ser. No. 18/821,454, filed Aug. 30, 2024, titled “REINFORCED PORCELAIN PANEL SYSTEM AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” now U.S. Pat. No. 12,247,403, issued Mar. 11, 2024, which claims priority to and the benefit of U.S. Provisional Application No. 63/676,400, filed Jul. 28, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/651,803, filed May 24, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/643,778, filed May 7, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/551,903, filed Feb. 9, 2024, titled “MAGNETIC SHOWER WALL

ACCESSORIES,” U.S. Provisional Application No. 63/549,820, filed Feb. 5, 2024, titled “PORCELAIN BOARD TILES,” U.S. Provisional Application No. 63/549,704, filed Feb. 5, 2024, titled “PORCELAIN BOARD FOR FABRICATORS OR DISTRIBUTORS,” and U.S. Provisional Application No. 63/541,981, filed Oct. 2, 2023, titled “SYSTEMS AND METHODS FOR PORCELAIN BOARD,” the disclosures of all of which are incorporated herein by reference in their entireties. U.S. Non-Provisional application Ser. No. 18/902,121 is also a continuation-in-part of U.S. Non-Provisional application Ser. No. 18/821,478, filed Aug. 30, 2024, titled “REINFORCED PORCELAIN PANEL SYSTEM AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” which claims priority to and the benefit of U.S. Provisional Application No. 63/676,400, filed Jul. 28, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/651,803, filed May 24, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/643,778, filed May 7, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/551,903, filed Feb. 9, 2024, titled “MAGNETIC SHOWER WALL ACCESSORIES,” U.S. Provisional Application No. 63/549,820, filed Feb. 5, 2024, titled “PORCELAIN BOARD TILES,” U.S. Provisional Application No. 63/549,704, filed Feb. 5, 2024, titled “PORCELAIN BOARD FOR FABRICATORS OR DISTRIBUTORS,” and U.S. Provisional Application No. 63/541,981, filed Oct. 2, 2023, titled “SYSTEMS AND METHODS FOR PORCELAIN BOARD,” the disclosures of all of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure generally relates to a reinforced porcelain panel product fabrication methods for enhanced structural protection. More specifically, the present disclosure relates to embodiments of a reinforced porcelain panel product fabrication methods for enhanced structural protection to reduce risk of chips or cracks from handling, installing, cutting, or transporting of the reinforced porcelain panel product.

BACKGROUND

(2) Porcelain slabs have various uses within residential and commercial locations. In particular, bathrooms, common spaces, such as living rooms and kitchens, lobbies, and other similar occupied spaces, may benefit from the aesthetic design of porcelain slabs. Furthermore, porcelain can be crafted, or engineered, to display a variety of designs, such as, for example, a marble appearance, a metallic appearance, or other similar features. While porcelain is an ideal aesthetic material for construction, among other benefits, porcelain, however, is a delicate and brittle material that is vulnerable to chips and cracks upon mishandling, such as during fabrication, shipping, and installation. Applicant has recognized that the implementation of proper handling training, tools, and techniques to move or install porcelain has not achieved a suitable protection from stress induced fractures of the porcelain from mishandling.

SUMMARY

(3) Applicant has recognized a need for enhanced protection of porcelain slabs from stress induced fractures during fabrication, installation, transport, handling, or cutting. Applicant has recognized a need for reducing a risk of stress induced fractures during fabrication, fabrication, installation, transport, handling, or cutting caused by mishandling, such as, for example, collisions of delicate porcelain slabs, which lead to loss of material, non-aesthetic repairs, and increasing the cost of construction. Embodiments of the present disclosure include, for example, a method to fabricate a reinforced porcelain panel product for renovations or new constructions, as would be understood by one skilled in the art. The method comprising includes applying an adhesive material to a top surface of a structural core board. The structural core board includes a foam-core material and has a

bottom surface opposite the top surface. The method further includes abuttingly contacting the structural core board to a bottom surface of a porcelain slab with the adhesive material therebetween such that regions of the top surface of the structural core board extend beyond outer peripheries of a first surface area of a top surface of the porcelain slab to define buffer regions, thereby to allow the buffer regions to be visible when viewing the reinforced porcelain panel product from a front view of a top surface of a reinforced porcelain panel product produced and to enhance protection from cracking when the reinforced porcelain panel product is handled. The structural core board has a second surface area larger than the first surface area of a top surface of the porcelain slab and a fracture toughness greater than the porcelain slab.

(4) In yet another embodiment, for example, the method includes pre-sizing the porcelain slab to have a thickness of about 3-millimeters to about 8-millimeters prior to applying the adhesive material to a top surface of a structural core board. Additionally, the method includes pressing the structural core board onto the porcelain slab after abuttingly contacting the structural core board to a bottom surface of a porcelain slab to spread the adhesive material thereon evenly therebetween, thereby to reduce air between the structural core board and to affix the structural core board onto the porcelain slab when in the adhesive material cures. The method further includes trimming outer peripheries of the structural core board such that the structural core board extends an equidistant distance beyond all outer peripheries of the first surface area of the porcelain slab by an overhang of about 0.125 inch to about 0.750 inches. Further, the adhesive material is applied by techniques using a spray, roller, bead applicators, or a combination thereof, and wherein the adhesive material includes a silyl-terminated polyether base resin.

(5) In still another embodiment, the fracture toughness of the structural core board resists a fracture after the porcelain slab is divided such that the structural core board remains intact when the porcelain slab is divided into two or more portions, so that the structural core board remains intact after the porcelain slab is divided into the two or more portions. The structural core board extends beyond outer peripheries of the first surface area of the porcelain slab by an overhang length within about 0.125 inch to about 0.750 inches evenly around the outer peripheries of the first surface area of the porcelain slab.

(6) In another embodiment, the foam-core material is formed by a technique using molding, pressing, layering, or the like, and includes a thermoplastic, wherein the thermoplastic consists of polyurethane, polycarbonate, polyphenylene oxide, polybutylene terephthalate, polyethylene terephthalate, or acrylonitrile butadiene styrene. The foam-core material has a young's modulus value lower than the porcelain slab and the young's modulus value of the foam-core material has a range of about 2.0 GPa to about 14.6 GPa.

(7) In an additional embodiment, the reinforced porcelain panel product has an impact resistance of at least a 30% greater than porcelain having a thickness of 6 mm or higher and a breaking strength of at least 40% greater than porcelain having a thickness of at least 6 mm. The foam-core material has a young's modulus value lower than the porcelain slab and the young's modulus value of the foam-core material has a range of about 2.0 GPa to about 14.6 GPa. Additionally, the structural core board extends beyond outer peripheries of the first surface area of the porcelain slab by an overhang length within about 0.125 inch to about 0.750 inches evenly around the outer peripheries of the first surface area of the porcelain slab.

(8) In another embodiment, the method further includes coating the bottom surface of the structural core board such that the bottom surface is substantially non-permeable, thereby to (a) enable use of a vacuum tool to position the reinforced porcelain panel product in various locations during fabrication, or (b) substantially waterproof the bottom surface.

(9) In still another embodiment, for example, the method further includes scoring the top surface of the porcelain slab with one or more scorelines positioned along one or more selected locations across the top surface, thereby to guide controlled breaks to cut the porcelain slab into a selected pattern when a radial force is exerted on or near the scorelines.

(10) In yet another embodiment, the method further includes (a) cutting the porcelain slab along the one or more scorelines to make one or more tiles of the reinforced porcelain panel product, (b) positioning the one or more tiles of the reinforced porcelain panel product in a selected orientation, and c) abuttingly affixing one or more tiles to a mounting surface adhesively.

(11) In yet another embodiment, the method further includes cutting the reinforced porcelain panel product to fit kitchen fixtures such that the reinforced porcelain panel product is laid on a kitchen structure to be used as a counter having the kitchen fixtures when installed.

(12) In another embodiment of the present disclosure, a method to fabricate a reinforced porcelain panel product for renovations or new constructions, as would be understood by one skilled in the art, includes pre-sizing a porcelain slab to have a thickness of about 3-millimeters to about 8-millimeters and applying an adhesive material to a top surface of a structural core board, the structural core board including a bottom surface opposite the top surface. The adhesive material is applied by a technique using a spray, a roller, a bead applicator, or a combination thereof. The method further includes abuttingly contacting the structural core board to a bottom surface of a porcelain slab with the adhesive material therebetween such that regions of the top surface of the structural core board extend beyond outer peripheries of a first surface area of the porcelain slab to define a buffer region, thereby to allow the buffer region to be visible when viewing a reinforced porcelain panel product produced from a front view of the reinforced porcelain panel product and to enhance an impact resistance of at least a 30% greater than porcelain having a thickness of 6 mm or higher from cracking or chipping when the reinforced porcelain panel product is handled. The buffer region includes a first pair of peripheral opposite side portions positioned substantially parallel with each other and a second pair of opposite side portions connected to and extending transverse to the first pair of peripheral opposite side portions also positioned substantially parallel to each other. The structural core board has a thickness greater than the porcelain slab and a young's modulus value lower than the porcelain slab. The top surface of the structural core board has a second surface area greater than a first surface area of the porcelain slab while the bottom surface of the structural core board is to be attached to a mounting surface when installed. The method continues with trimming outer peripheries of the structural core board such that the structural core board extends an equidistant distance beyond all outer peripheries of the first surface area of the porcelain slab.

(13) In an embodiment, the method includes scoring the top surface of the porcelain slab with one or more scorelines positioned along one or more selected locations across the top surface, thereby to guide controlled breaks to cut the porcelain slab into a selected pattern when a radial force is exerted on or near the scorelines. Additionally, the scoring is performed with a glass cutting wheel or a utility knife.

(14) In still another embodiment of the present disclosure, for example, a method to fabricate one or more reinforced porcelain panel products for renovations or new constructions, as would be understood by one skilled in the art, includes applying an adhesive material to a top surface of a structural core board. The structural core board includes a foam-core material, a bottom surface opposite the top surface, and a total surface area defined by summing surface areas from all surfaces of the structural core board. The method further includes positioning the structural core board such that top surface of a structural core board aligns with a preselected position of a bottom surface of a porcelain slab and abuttingly contacting the structural core board to the bottom surface of the porcelain slab with the adhesive material therebetween at the preselected position to produce a reinforced porcelain panel product. The structural core board has a thickness greater than the porcelain slab such that the total surface area of the structural core board is greater than the total surface area of the porcelain slab, thereby to enhance protection from cracking or chipping when the reinforced porcelain panel product is handled. The bottom surface of the structural core board (a) has a first surface plane substantially parallel to the top surface of the porcelain slab and (b) is to be attached to a mounting surface when installed. The method further includes pressing the

structural core board onto the porcelain slab to spread the adhesive material thereon evenly therebetween, thereby to reduce air between the structural core board onto the porcelain slab and to affix the structural core board onto the porcelain slab when in the adhesive material cures.

(15) In some embodiments, the adhesive material is applied by techniques using a spray, roller, bead applicators, or a combination thereof, and wherein the adhesive material includes a silyl-terminated polyether base resin.

(16) In other embodiments, outer peripheries of the structural core board substantially align with outer peripheries of the porcelain slab such that outer peripheries of the structural core board are substantially flush aligned with the outer peripheries of the porcelain slab.

(17) In an embodiment of the present disclosure, for example, a method to fabricate one or more reinforced porcelain panel products for renovations or new constructions, as would be understood by one skilled in the art, includes applying an adhesive material to a top surface of a structural core board. The structural core board includes a foam-core material, a bottom surface opposite the top surface, a total surface area defined by summing surface areas from all surfaces of the structural core board, and has a thickness up to 10-millimeters. The method further includes positioning the structural core board such that top surface of a structural core board aligns with a preselected position of a bottom surface of a porcelain slab. The porcelain slab has a thickness of about 3-millimeters to about 8-millimeters. The method further includes abuttingly contacting the top surface of the structural core board to the bottom surface of the porcelain slab with the adhesive material therebetween at the preselected position to produce a reinforced porcelain panel product. The structural core board has a thickness greater than the porcelain slab such that the total surface area of the structural core board is greater than the total surface area of the porcelain slab and a second surface area of the top surface of the structural core board is substantially equal to a first surface area of the top surface of the porcelain slab such that outer peripheries of the structural core board substantially align with outer peripheries of the first surface area of the porcelain slab, thereby to enhance protection from cracking or chipping when the reinforced porcelain panel product is handled. The bottom surface of the structural core board (a) has a first surface plane substantially parallel to the top surface of the porcelain slab and (b) is to be attached to a mounting surface when installed. The method further includes pressing the structural core board onto the porcelain slab to spread the adhesive material thereon evenly therebetween, thereby to reduce air between the structural core board onto the porcelain slab and to affix the structural core board onto the porcelain slab when in the adhesive material cures.

(18) In another embodiment, the adhesive material is applied by techniques using a spray, roller, bead applicators, or a combination thereof. Additionally, the adhesive material includes a silyl-terminated polyether base resin.

(19) In still another embodiment, for example, the structural core board includes two or more interlocking backing segments so that each of the two or more interlocking backing segments interlock to another of the two or more interlocking backing segments. Each of the two or more interlocking backing segments comprises a finger joint, a shiplap joint, or a dovetail joint each configured to adhesively interlock each of the two or more interlocking backing segments together.

(20) Aspects and advantages of these exemplary embodiments and other examples, are discussed in detail herein. Moreover, it is to be understood that both the foregoing information and the following detailed description provide merely illustrative examples of various aspects and embodiment and are intended to provide an overview or framework for understanding the nature and character of the claimed aspects and embodiments. Accordingly, these and other objects, along with advantages and features of the present disclosure, will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and may exist in various combinations and permutations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the embodiments of the present disclosure, are incorporated in and constitute a part of this specification, illustrate embodiments of the present disclosure, and together with the detailed description, serve to explain principles of the embodiments discussed herein. No attempt is made to show structural details of this disclosure in more detail than may be necessary for a fundamental understanding of the embodiments discussed herein and the various ways in which they may be practiced. According to common practice, the various features of the drawings discussed below are not necessarily drawn to scale. Dimensions of various features and elements in the drawings may be expanded or reduced to illustrate embodiments of the disclosure more clearly.

(2) FIG. 1 is perspective view of an embodiment of a reinforced porcelain panel product, according to one embodiment of the present disclosure.

(3) FIG. 2A is a sectional view of the reinforced porcelain panel product taken along the lines of FIG. 2A of the perspective view of an embodiment of a reinforced porcelain panel product of FIG. 1, according to another embodiment of the present disclosure.

(4) FIG. 2B is a front view of an embodiment of a reinforced porcelain panel product with a buffer region, according to one embodiment of the present disclosure.

(5) FIG. 2C is a perspective view of an embodiment of a reinforced porcelain panel product without a buffer region, according to one embodiment of the present disclosure.

(6) FIG. 3 is an open perspective view of an embodiment of a reinforced porcelain panel product, according to one embodiment of the present disclosure.

(7) FIG. 4A is an exploded view an embodiment of the reinforced porcelain panel product having interlocking backing segments in an interlocking configuration, according to one embodiment of the present disclosure,

(8) FIG. 4B is an exploded view an embodiment of a reinforced porcelain panel product having a single structural core board, according to one embodiment of the present disclosure.

(9) FIG. 5A is an exemplary handling embodiment of reinforced porcelain panel products using a lifting clamp, according to one embodiment of the present disclosure.

(10) FIG. 5B is a sectional view of the compressed reinforced porcelain panel product by the lifting clamp taken along the lines of 5B of the perspective view of an embodiment of a reinforced porcelain panel product of FIG. 5A, according to another embodiment of the present disclosure.

(11) FIG. 6A is an exemplary transport embodiment of stacking reinforced porcelain panel products on a pallet for transporting the reinforced porcelain panel products, according to one embodiment of the present disclosure.

(12) FIG. 6B is an exemplary transport embodiment of the pallets of reinforced porcelain panel products transported with freight shipping, according to one embodiment of the present disclosure.

(13) FIG. 6C is an exemplary transport embodiment of the pallets of reinforced porcelain panel products without buffer region transported with freight shipping, according to one embodiment of the present disclosure.

(14) FIG. 6D is an exemplary transport embodiment of the reinforced porcelain panel products being handled to be loaded onto a delivery truck, according to one embodiment of the present disclosure.

(15) FIGS. 7A-E are exemplary cutting operations of the reinforced porcelain panel product, according to one embodiment of the present disclosure.

(16) FIG. 8 is an exemplary installation of the reinforced porcelain panel product, to one embodiment of the present disclosure.

(17) FIGS. 9A-D are exemplary installations of the reinforced porcelain panel product in a

bathroom setting, according to various embodiments of the present disclosure.

(18) FIG. 10 is an exemplary installation of a trim piece constructed from a reinforced porcelain panel product, according to one embodiment of the present disclosure.

(19) FIG. 11 is an exemplary installation of a countertop constructed from a reinforced porcelain panel product in a kitchen with a cutout showing the structural core board of the installed countertop, according to one embodiment of the present disclosure.

(20) FIG. 12 is an exemplary installation of a tiled art piece displayed on a wall constructed from a reinforced porcelain panel product, according to one embodiment of the present disclosure.

(21) FIGS. 13A and 13B are a flow chart of a method of fabricating a reinforced porcelain panel product for commercial or residential renovations or new constructions, according to one embodiment of the present disclosure.

(22) FIG. 14 is a flow chart of a method of transporting and using a reinforced porcelain panel product, according to one embodiment of the present disclosure.

(23) FIG. 15 is a flow chart of a method for lifting a reinforced porcelain panel product using a lifting clamp, according to one embodiment of the present disclosure.

(24) FIGS. 16A and 16B are perspective views of a reinforced porcelain panel product kit, according to two embodiments of the present disclosure.

(25) FIG. 16C is a schematic view of a reinforced porcelain panel product kit, according to one embodiment of the present disclosure.

(26) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated into other embodiments without further recitation.

DETAILED DESCRIPTION

(27) The present disclosure describes various embodiments of reinforced porcelain panel products and associated methods for enhanced structural protection to reduce risk of chips or cracks from a handling of the reinforced porcelain panel products by protection of the porcelain slab with a structural core board. Porcelain slabs have various utility in the construction and decoration industry. Large porcelain slabs may be designed, textured, decorated, or engineered for specific aesthetic appealing walls. However, large porcelain slabs are challenging to transport, fabricate, and install as expert, or routine, handling methods may result in broken, fractured, chipped, or cracked porcelain slabs attributed to mishandling of the fragile porcelain slab. These mishandling failures necessitate repairs that increase cost of construction projects, repair solutions that may be undesirable to the purchaser, and/or lost productivity of fabrication and installation.

(28) The below description may use the phrases “in certain embodiments,” “in various embodiments,” “in an embodiment,” “in one embodiment, or “in example,” which may each refer to one or more of the same or different embodiment. Furthermore, the terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. The term “plurality” as used herein refers to one or more items or components. The terms “about” or “approximately” are defined as being close to as understood by one of ordinary skill in the art. In one non-limiting example, these terms are defined to be within 10 percent (%), preferably within 5%, more preferably within 1%, and most preferably within 0.5%.

(29) The terms “removing,” “removed,” “reducing,” “reduced,” or any variation thereof, when used in the claims and/or the specification includes any measurable decrease of one or more components in a mixture to achieve a desired result. The use of the words “a” or “an” when used in conjunction with any of the terms “comprising,” “including,” “containing,” or “having,” in the claims or the specification may mean “one,” but it is also consistent with the meaning of “one or more,” “at least one,” and “one or more than one.” The terms “wt. %,” “vol. %,” or “mol. %” refers to a weight, volume, or molar percentage of a component, respectively, based on the total weight, the total volume of material, or total moles, which includes the component. In a non-limiting example, 10

grams of a component in 100 grams of the material is 10 wt. % of the component.

(30) Generally, fabricated porcelain material may include a ceramic made by heating porcelain-creating raw materials to temperatures commonly between about 1,200 degrees Celsius and about 1,400 degrees Celsius. Porcelain-creating raw materials may include kaolin, such as kaolinite, feldspar, ball clay, glass, bone ash, steatite, quartz, petuntse and alabaster. Porcelain may include natural patterning, such as veins, which offers a variety of decorative purposes, such as art displays. Further, porcelain may be manipulated to display a variety of finishes or designs, for example, polished, natural, honed, marble look, wood-look, metal-look, concrete-look, fabric-look and may possess grains therein to exhibit types of textures created by varied processes. Porcelain may also be etched, or engraved, to include descriptive text for signage or other functional uses. Therefore, porcelain is an attractive material for wall cladding, kitchen countertops, tabletops, fireplaces, shower walls, floors, pools, and/or other construction uses due to its ability to be manipulated with a variety of designs or textures.

(31) Porcelain may be fabricated in sheets of porcelain referred herein as porcelain slabs. The porcelain slabs described herein can include various types of porcelain materials, types including sintered stone, such as neolith, or ultra compact porcelain, such as florim, having various grades and/or qualities. One use of porcelain includes fabricating porcelain large panels for use in occupied spaces, such as, for example, lodgings, hotels, houses, new space construction, and/or renovations of the former. However, as discussed above, porcelain slabs are large, thin, and delicate, and therefore, the panels are susceptible to chip or crack when used by fabricators, stone masons, or the like, as compared to other more tolerant materials, such as quartz, or granite, leading to a reduction of the use of porcelain slabs in construction projects to reduce risk of mishandling in the fabrication cutting and/or installation process.

(32) The following disclosure is related to thin and large porcelain slabs, commonly referred to as “thin large format porcelain” amongst those skilled in the art. However, the solution provided to the challenges plaguing thin large format porcelain herein may be also provided for various sizes of porcelain slabs to reduce the risk of stress induced fractures from transportation and mishandling. Further, the following disclosure refers to “panel(s)” as sheets, or portions of slabs of porcelain material that are independent, or stand alone, and further refers to “board(s)” as integrated, or fabricated materials that are united to provide structure, such as a structural core board having an internal structural core. Furthermore, the following disclosure may refer to “assembly(ies),” as integrated materials that are adhesively united, such as a porcelain slab and a structural core board to define a reinforced porcelain panel product being produced under the trademark “MightySlab™.”

(33) FIG. 1 is perspective view of an embodiment of a reinforced porcelain panel product **100** according to one embodiment of the present disclosure. FIG. 1 illustrates a reinforced porcelain panel product **100** including a porcelain slab **102** and a structural core board **130** positioned thereon. The porcelain slab **102** has a top surface **104** and a bottom surface **106** (as shown in FIG. 2A) positioned opposite of the top surface **104**. The top surface **104** has a surface area **108**. In one embodiment, the bottom surface **106** has a surface area that substantially matches the surface area **108** of the top surface **104**. In one embodiment, the porcelain slab **102** is rectangular, although various shapes are contemplated.

(34) The structural core board **130** has a top surface **134** and a bottom surface **136** positioned opposite of the top surface **134**. In one embodiment, top surface **134** and a bottom surface **136** each have a plane that are substantially parallel to each other. The top surface **134** has a surface area **138**. In one embodiment, the bottom surface **136** has a surface area that substantially matches the surface area **138** of the top surface **134**. In one embodiment, the structural core board **130** is rectangular, although various shapes are contemplated as discussed below. The structural core board **130** may be constructed from a foam-core material such as a variety of structural foams, foams, or other foam-core materials with similar foam strength, weight, and/or structural

characteristics. In some examples, the foam-core material may be formed by techniques using molding, pressing, layering, or the like, as would be understood by one skilled in the art. In another embodiment, the structural core board **130** includes polyethylene terephthalate ("PET") foam. In some embodiments, the PET foam may be mixed with glass fibers or carbon fiber. In one embodiment, the bottom surface **136** of the structural core board **130**, may be laminated, thereby to have a smooth surface for use on air suction tables for positioning, for example, a CNC table, as would be understood by those skilled in the art. In one embodiment, the PET foam is closed-cell PET foam formulated to provide enhanced mechanical properties from, for example, recyclable material such as plastic bottles. Furthermore, PET foam mechanical properties have exhibited significant structural resilience in other industry uses such as, the wind industry on turbine blades, the marine industry on decks and hulls of vessels, the railroad industry on train body sidewalls and flooring supports, and the aerospace industry on plane structural periphery supports. PET foam is temperature tolerant up to about 150 degrees Celsius. In some embodiments, PET foam has a young's modulus of about 2.0 GPa to about 14.6 GPa and a fracture toughness ranging from about 3 MPa-m.^{sup.1/2} to about 9.5 MPa-m.^{sup.1/2}. In comparison, porcelain has a young's modulus of about 67 GPa to about 150 GPa and a fracture toughness of about 2.0 MPa-m.^{sup.1/2}. Therefore, PET foam is less brittle and more fracture resistant as compared to the mechanical properties of porcelain. Furthermore, the structural core board **130** may weigh approximately 0.018 pounds per square foot. For example, suitable commercially available PET foam includes recyclable polymer foams, such as AIREX® T92. Conventionally, alternate 0.375 inches thick (or about 10-millimeters) support products for stone structural support, such as honeycomb plates, is about 3.5 to about 4 times the weight per square foot as compared to the structural core board **130** embodiment disclosed herein. As an example of the total weight of the reinforced porcelain panel product **100**, including the structural core board **130**, a 6-millimeter-thick reinforced porcelain panel product **100** may weigh about 150 pounds as compared to a similarly sized 2-centimeter thick quartz panel which may weigh about 800 pounds. Therefore, weight reduction may be advantageous in weight sensitive applications, such as, for example, yachts or other weight sensitive construction.

(35) FIG. 2A is a sectional view of the reinforced porcelain panel product **100** taken along the lines of FIG. 2A of the perspective view of an embodiment of the reinforced porcelain panel product **100** of FIG. 1, according to another embodiment of the present disclosure. FIG. 2A illustrates the reinforced porcelain panel product **100** including the porcelain slab **102**, the structural core board **130** positioned thereon, and an adhesive material **250** positioned between the porcelain slab **102** and the structural core board **130**. The structural core board **130** has a thickness **240** and the porcelain slab **102** has a thickness **210**. In one embodiment, for example, the structural core board **130** may be pre-sized to have a thickness **240** up to about 0.375 inches thick (or about 10-millimeters, such as less than 10-millimeters, or up to about 10-millimeters), although dimensions may be adjusted for specific scenarios of handling. In another embodiment, for example, the porcelain slab **102** may be pre-sized to have a thickness **210** of about 3-millimeter to about 8-millimeter thick, such as about 3.5-millimeter to about 7-millimeter, such as about 3.5-millimeter to about 6-millimeter, such as about 3.5-millimeter, such as about 6-millimeter, or such as about 7-millimeter. In yet another embodiment, for example, the thickness **240** of the structural core board **130** is greater than the thickness **210** of the porcelain slab **102**.

(36) The adhesive material **250** unites the bottom surface **106** of the porcelain slab **102** to the top surface **134** of the structural core board **130** such that the porcelain slab **102** and the structural core board **130** are abutting connected and physically contacting. In one embodiment, the adhesive material **250** permanently bonds the porcelain slab **102** and the structural core board **130** together. In one embodiment, the adhesive material **250** is evenly spread across the top surface **134** of the structural core board **130** to promote an even spread of adhesive material **250** across the bottom surface **106** of the porcelain slab **102** when installed, as shown in FIG. 3. In one embodiment, the

adhesive material **250** may be bead across the top surface **134** of the structural core board **130** and troweled prior to engaging with the bottom surface **106** of the porcelain slab **102**, as would be understood by those skilled in the art. In yet another embodiment, the structural core board **130** may be back buttered and further vibrated, or pressed onto the porcelain slab **102**, to promote adhesive material **250** coverage across the area of the bottom surface **106** of the porcelain slab **102** to reduce air pockets which advantageously provides tight sealing and support across the mated surfaces, thereby reducing risk of chips, fractures, or other damage, to the porcelain slab **102** at unsupported sites between the mated surfaces when installed. In still another embodiment, the adhesive material **250** may be applied as a spray across the top surface **134** of the structural core board **130**. In another embodiment, the adhesive material **250** may be applied with a roller, or similar brushing technique across the top surface **134** of the structural core board **130**. In some embodiments, the adhesive material **250** may be alternatively applied to the bottom surface **106** of the porcelain slab **102**. The adhesive material **250** may selected from a variety of specialty adhesives, laminates, glues, or similar joining applications that are suitable for porcelain adhesion on to the material selected for the structural core board **130**. In one embodiment, the adhesive material **250** contains a polymer. In another embodiment, the adhesive material **250** contains low-VOC Silyl-Terminated Polyether (STPE) base resin, such as provided by KANEKA MS Polymers™.

(37) FIG. 2B is a front view of an embodiment of a reinforced porcelain panel product **100** with a buffer region **260**, according to one embodiment of the present disclosure. The front view of FIG. 2B shows an embodiment, for example, of the porcelain slab **102** having a length **212** of about 74 inch, about 96 inch, about 106 inch, about 109 inch, about 120 inch, or about 126 inch and a width **214** of about 31 inch, about 31.5 inch, about 48 inch, about 60 inch, or about 63 inch, although various sizes may be available, as will be understood by those skilled in the art. For example, commercial standard porcelain slabs **102** are produced by the fabricator in sizes of about 63 inch by about 126 inch, about 60 inch by 126 inch, about 60 inch by about 120 inch, or about 48 inch by about 109 inch. In another embodiment, for example, the porcelain slab **102** may be fabricated or cut to varied sizes, such as, but not limited to, about 31.5 inch by 74 inch, about 31.5 by 96 inch, about 31.5 inch by 106, about 48 inch by 96 inch, about 48 inch by 108 inch, about 60 inch by 74 inch, about 60 inch by 96 inch, about 60 inch by 106 inch, about 60 inch by 120 inch about 60 inch by 126 inch, or about 63 inch by about 126 inch. The largest of the porcelain slab **102** sizes is particularly used for delicate shower installations. In one embodiment, the structural core board **130** has a length **242** of less than about 75.5 inch, 97.5 inch, 107.5 inch, 121.5 inch, or less than about 127.5 inch and a width **244** of less than about 32.5 inch, 33 inch, 49.5 inch, 61.5 inch, or less than about 64.5, although various sizes are contemplated. In another embodiment, the structural core board **130** has a length **242** greater than the length **212** by about 0.250 inch to about 1.500 inch, such as about 0.350 inch to about 1.000 inch, such as about 0.400 inch to about 0.860 inch, such as about 0.500 inch to about 0.760 inch, such as about 0.600 inch to about 0.660 inch, such as about 0.625 inch, although various sizes are contemplated.

(38) Furthermore, FIG. 2B shows the structural core board **130** larger than the porcelain slab **102**, thereby to create a buffer region **260** defined by the visible difference of size between the structural core board **130** and the porcelain slab **102**. Stated differently, the surface area **138** of the structural core board **130** has a greater surface area than the surface area **108** of the porcelain slab **102** such that the buffer region **260** visibly borders the porcelain slab **102** on the top surface **134** of the structural core board **130** from a front view, as illustrated in FIG. 2B. Stated in another way, the regions of the top surface **134** of the structural core board **130** substantially extend beyond outer peripheries of the first surface area **108** of the porcelain slab **102** and define a buffer region **260** so that a combination of the porcelain slab **102** and the structural core board **130** define the reinforced porcelain panel product **100**. Furthermore, the buffer region **260** includes a first pair of peripheral opposite side portions, a length overhang **246**, positioned substantially parallel with each other and

a second pair of peripheral opposite side portions, a width overhang **248**, connected to and extending transverse to the first pair of peripheral opposite side portions also positioned substantially parallel to each other. The size difference and placement of the porcelain slab **102** onto the structural core board **130**, or vice versa, creates the length overhang **246** and the width overhang **248** that partially dimensionally defines the buffer region **260**.

(39) In one embodiment, for example, the length overhang **246** may be about 0.313+/-0.001 inch and the width overhang **248** may be about 0.313+/-0.001 inch. In another embodiment, for example, the length overhang **246** and the width overhang **248** are substantially equivalent having an overhang of about 0.125 inch to about 0.750 inch, such as about 0.175 inch to about 0.500 inch, such as about 0.200 inch to about 0.430 inch, such as about 0.250 inch to about 0.380 inch, such as about 0.300 inch to about 0.330 inch, such as about 0.313+/-0.001 inch, or such as less than about 0.750+/-0.001 inches, such as less than about 0.500 inches, such as less than about 0.43 inches, such as less than about 0.380 inches, or such as less than about 0.330 inches. Stated differently, the structural core board may extend an equidistant distance beyond all outer peripheries of the porcelain slab. In some embodiments, the structural core board **130** may be trimmed, after positioning onto the porcelain slab **102**, to possess the aforementioned length overhang **246** and the width overhang **248** distances.

(40) The length overhang **246** and the width overhang **248** of the structural core board **130** are particularly advantageous as the defined buffer region **260** has been empirically found to protect the porcelain slab **102** from chips, cracks, fractures, or the like, as will be understood by those skilled in the art, from mishandling of the reinforced porcelain panel product **100** during transport or installation. For example, the porcelain slab **102**, as discussed above, is large, thin, and fragile such that bumps or an unassisted lift may cause stresses, such as shear, bend, torsion, torque, and the like, which may lead to a failure of the porcelain slab to remain intact, undamaged, or not chipped. However, the length overhang **246** and the width overhang **248** have empirically been found to significantly reduce the likelihood of fractures, and the like, from inadvertent drops, such as falls directly onto the bottom surface **136** of the structural core board **130** drops, corner bumps, or other similar collisions with another hard surface, such as a floor, crate, another panel, or other construction hazards that commonly occur with transportation and installation process of the porcelain slabs. Furthermore, the inventors discovered that overhangs exceeding the above disclosure are prone to failure by breaking off the porcelain slab **102** and thereby to expose the porcelain slab **102** to risk of stress induced damage. Similarly, the inventors discovered that overhangs not exceeding the lower values of the above disclosure are likely to expose the porcelain slab **102** to stress induced damage by not providing adequate protection against inadvertent drops, bumps, or other similar collisions on the sides. Therefore, the length overhang **246** and the width overhang **248** of the structural core board **130** are critical to the protection of the porcelain slab **102** to stress induced damage related to inadvertent drops, bumps, or other similar collisions with another hard surface that commonly occurs with transportation and installation process of the porcelain slabs.

(41) FIG. 2C is a perspective view of an embodiment of a reinforced porcelain panel product **100** without a buffer region **260** as compared to FIG. 2B, according to one embodiment of the present disclosure. FIG. 2C illustrates an embodiment of the reinforced porcelain panel product **100** where the structural core board **130** and the porcelain slab **102** are substantially the same size at their outer peripheries. However, in some embodiments, as discussed above, the thickness of the structural core board **130** is greater than thickness of the porcelain slab **102** such that the total surface area, defined by summing surface areas from all surfaces of the structural core board **130**, is greater than the total surface area of the porcelain slab **102**, defined by summing surface areas from all surfaces of the of the porcelain slab **102**. In some embodiments, the porcelain slab **102** may have sides, about the outer peripheries, that are flush, or substantially flush with the structural core board **130**. The porcelain slab **102** may be dimensionally sized as discussed in FIG. 2B above,

including length, width, and thickness. Similarly, the structural core board **130** may also possess the length and width dimensions discussed in FIG. 2B pertaining to the porcelain slab **102**.

(42) The embodiment of FIG. 2C may be fabricated without the buffer region **260** of FIG. 2B for various reasons. In one embodiment, an installation may require the buffer region **260** of FIG. 2B to be absent, thereby encouraging fabricators to ship the reinforced porcelain panel product **100** as shown in FIG. 2C. As will be discussed below, trim pieces may be added to edges of the reinforced porcelain panel product **100** to conceal the union between the structural core board **130** and the porcelain slab **102**, such as adhesive **250**. The embodiment of FIG. 2C may be utilized in delicate transportations wherein the transport of the reinforced porcelain panel product **100** may not experience substantial turbulence. As illustrated, the structural core board **130** provides protection from stresses applied onto the top surface **104** of the porcelain slab **102**, such as shear, bend, torsion, torque, and the like, which may lead to a failure of, or damage to, the porcelain slab **102**.

(43) FIG. 3 is an open perspective view of an embodiment of a reinforced porcelain panel product **100**, according to one embodiment of the present disclosure. FIG. 3 illustrates a preinstallation view of the porcelain slab **102** to be attached to the structural core board **130** to construct the reinforced porcelain panel product **100**. The porcelain slab **102** has a porcelain length side **304** and a porcelain width side **306**. Similarly, the structural core board **130** has a backing length side **334** and a backing width side **336**. In one embodiment, the porcelain slab **102** and the structural core board **130** are centrally aligned along alignment line **300** such that a center **302** of the porcelain slab **102** and a center **332** of the structural core board **130** contact when installed. In one embodiment, the porcelain length side **304** is substantially parallel to the backing length side **334**. In one embodiment, the porcelain width side **306** is substantially parallel to the backing width side **336**. The alignment of the porcelain slab **102** and the structural core board **130**, as discussed above, is advantageous as the buffer region **260** is uniform across the length, the width, or both, of the reinforced porcelain panel product **100** allowing the buffer region **260** to provide equal protection from contact against hard surfaces all around the porcelain slab **102**. Further, a uniform buffer region **260** provides a stone mason, or a fabricator, confidence when handling the reinforced porcelain panel product **100** as no one side is more exposed to potential damage.

(44) Foams commonly used for the structural core board **130** may be recycled polymeric foams, but others could be used as well. Structural foams commonly include a thermoplastic or a thermoset. In one embodiment, a commonly used thermoplastic includes, for example but not limited to, polyurethane, polycarbonate, polyphenylene oxide, polybutylene terephthalate, polyethylene terephthalate, and acrylonitrile butadiene styrene, although other materials are contemplated. In one embodiment, a chemical blowing agent may be used to promote foam expansion. In one embodiment, the foam may be a rigid polymeric foam including polyethylene terephthalate. The foam may exhibit desired mechanical properties including increased strength and/or low-weight characteristics, as discussed above. In some embodiments, the structural core board **130** may be chemically stable, exhibit good adhesion and fatigue strength, significantly low water absorption, among other features rendering the structural core board **130** desirable for the reinforced porcelain panel product.

(45) FIG. 4A is an exploded view an embodiment of the reinforced porcelain panel product **400** having interlocking backing segments, or structural core boards, in an interlocking configuration, according to one embodiment of the present disclosure. The reinforced porcelain panel product **400** may include the porcelain slab **102** and multiple interlocking backing segments **430a**, **430b**, and **430c**, although more or fewer interlocking backing segments are contemplated to support the porcelain slab **102**. The interlocking backing segments **430a**, **430b**, and **430c**, together define the structural core board **130**. In one embodiment, the dimensions of the interlocking backing segments **430a-430c** individually are less than the dimension of structural core board **130** discussed in FIGS. 1-4A. However, the embodiments using the interlocking backing segments **430a-430c** enable porcelain slabs **102** to be fabricated at sizes larger than presently fabricated as the segments may be

combined to fit larger thin large format porcelain. Stated differently, thin large format porcelain may be produced in sizes greater than presently available due the interlocking backing segments **430a-430c** providing tailored sizes equivalent to the purposes of the structural core board **130**, as discussed above. In some embodiments, the interlocking backing segments **430a-430c** are about 4 foot wide each wherein three backing segment panels may be combined and cut to construct one large panel, for example, a size of about 63.625 inch by about 126.625 inch.

(46) Each of the interlocking backing segments **430a-430c** may have one or more ends **432** for each of the interlocking backing segments **430a-430c**. Each of the one or more ends **432** have an interlocking configuration to connect the interlocking backing segments **430a-430c** to each other, as desired. In one embodiment, a method to fabricate the interlocking backing segments **430a-430c** may include fabricating the interlocking configurations onto the interlocking backing segments **430a-430c**, as discussed below, for protective use on the reinforced porcelain panel product **400**. As illustrated, the one or more ends **432** each have a shelf **435** configured to receive the opposite shelf **435** such that the shelf **435** is positioned on the opposite shelf **435** and joined to marry at least two of the interlocking backing segments **430a-430c**. In one embodiment, the one or more ends **432**, or interlocking configuration, may be an overlay joint, such as a finger, shiplap, or a dovetail joint, among other types of overlay joints, configured to join or fasten the interlocking backing segments **430a-430c** together. In one embodiment, adhesives may be used to join and secure one or more ends **432** together. In one embodiment, the interlocking backing segments **430a-430c** may be fabricated with adhesive tape positioned on mating surfaces of the one or more ends **432**, for example on the shelf **435**, to bond the shelf **435** of one backing segment to the opposite shelf **435** of another backing segment. In one embodiment, the adhesive tape may be covered with a peelable cover to expose the tape. In one embodiment, the interlocking backing segments **430a-430c** may be fabricated with porcelain-compliant adhesive tape positioned on the bottom surface **434** of each of the interlocking backing segments **430a-430c** to adhere to the bottom surface **106** of the porcelain slab **102**, thereby to construct the reinforced porcelain panel product **400**. Further, some of the interlocking backing segments, such as interlocking backing segments **430a** and **430c** may have termination ends **440** that are not configured to interlock with another backing segment but rather are configured to provide the appropriate overhang, as discussed above.

(47) In another embodiment, an interlocking configuration may be placed on the periphery edges of the reinforced porcelain panel product such that the structural core board comprises one or more interlocking edges configured to interlock with a structural core board of another reinforced porcelain panel product such that two or more reinforced porcelain panel products are interlockable and positionable adjacent each other. Stated differently, the peripheral outer edges of the structural core board may possess interlocking configurations that may be utilized to connect two or more reinforced porcelain panel products together when being installed. This feature is advantageous to reduce the amount of waste or structural core board cut by a worker when installing adjacent reinforced porcelain panel products. Further the interlocking configurations on the periphery edges of the reinforced porcelain panel product may provide adequate spacing for some installation embodiments.

(48) FIG. 4B is an exploded view an embodiment of a reinforced porcelain panel product **400** having a single structural core board **130**, according to one embodiment of the present disclosure. The structural core board **130**, as illustrated, may be positioned to be place on the porcelain slab **102** such that each side that are flush, or substantially flush, with the structural core board **130**, as similarly discussed in FIG. 2C. In one embodiment, a bottom surface **106** of the porcelain slab **102** may be covered with adhesive **250**. The structural core board **130** may be positioned, via line **438**, onto the bottom surface **106** of the porcelain slab **102** to fabricate reinforced porcelain panel product **400**. As discussed above, some installations may require a reinforced porcelain panel product **400** without a buffer region **206**. In particular, transport of the reinforced porcelain panel product **400** that lay, or stack, the reinforced porcelain panel product **400** may not demand the

buffer region **260** of FIG. 2B for protection against, for example, shear, bend, torsion, torque, and the like, as the structural core board **130** protects the porcelain slab **102** where adhesively attached. Unlike the embodiment of the reinforced porcelain panel product **400** containing the buffer region **260**, the reinforced porcelain panel product **400** without the buffer region **260** may be suitable for delicate handling and transportation scenarios.

(49) FIG. 5A is an exemplary handling embodiment for moving reinforced porcelain panel products **100** using a lifting clamp, according to one embodiment of the present disclosure. Fabricators use a variety of tools to handle stone panels. For example, air tables may be used to maneuver, or position, a laid down panel on, for example, a CNC machine. Other tools also include portable a-frames, such as shown in FIG. 6D but on caster wheels. Similarly, individual suction cups **526** may be used to provide handles for which workers **522** may move the panel. Another tool often used in a fabrication of stone is a lifting clamp **502** as shown in FIG. 5A. The lifting clamp **502** may be attached to an overhead rail system (not shown) by the clamp link **504**. The clamp link **504** may be a chain or similar device suitable for linkage between the overhead rail system and the lifting clamp **502**. In one embodiment, the lifting clamp **502** compresses the positioned reinforced porcelain panel product therein to be lifted and moved, for example, to a different location for further fabrication. In one embodiment, the lifting clamp **502** may lift the reinforced porcelain panel product **100** in a substantially vertical manner. In another embodiment, the weight of the reinforced porcelain panel product **100** may cause the lifting clamp **502** to angularly lift the reinforced porcelain panel product **100** for movement to the desired location. Conventionally, fabricators have grown accustomed to using the lifting clamp **502** on stone panels, such as, for example, quartz, granite, travertine, marble, and the like fracture resistant stone panels. However, fabricators have avoided using the lifting clamp **502** on porcelain materials as porcelain slabs have routinely failed, for example, by fracturing, shattering, or chipping, when being clamped and/or lifted by lifting clamp **502**. The reinforced porcelain panel product **100** of this disclosure, however, is able to be lifted or positioned by use of a lifting clamp **502**. The structural core board **130** of the reinforced porcelain panel product **100** advantageously provides stability and structural strength to protect against the forces exerted on the porcelain slab **102** by use of the lifting clamp **502** from fracturing, shattering, or chipping the porcelain slab **102**. Therefore, fabricators may advantageously apply their routine stone panel handling techniques to the reinforced porcelain panel product **100** embodiments, discussed above, without further special porcelain handling training and with a substantially reduced risk of product loss due to broken, fractured, chipped, or cracked porcelain slabs attributed to mishandling of the fragile porcelain slab.

(50) FIG. 5B is a sectional view of the compressed reinforced porcelain panel product by the lifting clamp taken along the lines of 5B of the perspective view of an embodiment of a reinforced porcelain panel product of FIG. 5A, according to another embodiment of the present disclosure. FIG. 5B illustrates a pair of movable jaws **552** within an interstitial space **550** of the lifting clamp **502**. In one embodiment, the interstitial space **550** is configured to open or close by an activation of the lifting clamp **502** to actuate expanding or narrowing the interstitial space **550** with the movable jaws **552**. Stated differently, the interstitial space **550** reduces and is eliminated as the movable jaws **552** close so as to clamp and secure, for example, the reinforced porcelain panel product **100**. Conversely, the interstitial space **550** may increase as the movable jaws **552** open to for example, release the reinforced porcelain panel product **100**. In one embodiment, the movable jaws **552** provide sufficient inward force to crush conventional unsupported porcelain. In other embodiments, the movable jaws **552** may not break the conventional unsupported porcelain, however, the conventional unsupported porcelain may still break upon a lifting of the porcelain by the lifting clamp. The present disclosure of the reinforced porcelain panel product **100** advantageously provides protection against breakage, chipping, and/or cracking of the porcelain slab **102** when using a lifting clamp **502**. In one embodiment, the structural core board **130** of the reinforced porcelain panel product **100** deforms to provide 1) a grip for the lifting clamp, and 2) an

ability to withstand the compression of the movable jaws **552** of the lifting clamp **502** to enable the lift. As illustrated, the structural core board **130** absorbs the compression power from the movable jaws **552** of the lifting clamp **502** to provide the protection to the porcelain slab **102**, as discussed, and the ability to move the reinforced porcelain panel product **100** to a desired location. Therefore, again, fabricators may advantageously apply their routine stone panel handling techniques, such as using a lifting clamp **502**, to the reinforced porcelain panel product **100** embodiments with a substantially reduced risk of product loss due to broken, fractured, chipped, or cracked porcelain slabs attributed to mishandling of the fragile porcelain slab.

(51) FIG. **6A-6D** are exemplary transport embodiments for transporting the reinforced porcelain panel products **100**. In particular, FIG. **6A** is an exemplary transport embodiment of stacking reinforced porcelain panel products on a pallet for transporting the reinforced porcelain panel products **100**, according to one embodiment of the present disclosure. FIG. **6B** is an exemplary transport embodiment of the pallets of reinforced porcelain panel products transported with freight shipping, according to one embodiment of the present disclosure. FIG. **6C** is an exemplary transport embodiment of the pallets of reinforced porcelain panel products without buffer regions transported with freight shipping, according to one embodiment of the present disclosure. FIG. **6D** is an exemplary transport embodiment of the reinforced porcelain panel products being handled to be loaded onto a delivery truck, according to one embodiment of the present disclosure. It is to be understood the term “stacking” is not intended to be limited to a vertical orientation, but rather is used to define a layering of two or more panels in either a vertical, horizontal, or angled orientation wherein the weight of the stacking is absorbed by the structural core board **130** of the reinforced porcelain panel product **100** discussed herein,

(52) FIG. **6A** illustrates a pallet of reinforced porcelain panel products **600** as an exemplary stacking of the reinforced porcelain panel products onto a pallet **602** for shipping for example, from a manufacturer to a fabricator. The reinforced porcelain panel products in FIG. **6A** may be the reinforced porcelain panel product **100**, as discussed above. In one embodiment, each of the reinforced porcelain panel products **100** may contact the stacked adjacent reinforced porcelain panel product **100**, such that no space is between the boards.

(53) FIG. **6A** provides a visualization of how the discussed panels are handled and possibly be damaged as a result from the transportation of the reinforced porcelain panel products **100**. For example, a fabricating site may produce the porcelain slabs **102** and attach the structural core board **130**, or interlocking backing segments **430a-c**, to create a reinforced porcelain panel product. Each of the boards must then be moved to the location of sale, fabrication, or the location of installation. One method of transporting the reinforced porcelain panel products **100**, **400** may be by a transport device, such as a pallet **602**, and freight shipping **610** as shown in FIGS. **6B** and **6C**. FIGS. **6B** and **6C** are similar illustrations however, the reinforced porcelain panel products **100**, **400** may or may not possess the buffer region **260**, as shown in FIG. **6B** and absent in FIG. **6C**. Each of the pallet of reinforced porcelain panel products **600** are subjected to various stresses including manufacturer, or fabricator, handling, pressure points from the straps used to tie down the boards onto the pallet **602**, loading **614** onto the freight truck **612** by, for example, a forklift, road vibrations or bumps from freight shipping **610**, and unloading to a fabrication or distribution site (as shown in FIG. **6D**). For example, conventional fabricators of porcelain slabs may chip or significantly crack unsupported porcelain slabs, when sawed, drilled, or when lifting the panels as discussed in relation with FIGS. **5A** and **5B**. Conventional fabrication shops may use heavy equipment, such as the lifting clamp **502**, to lift heavy panels that may not be sufficiently delicate to handle and carry thin panels, such as the porcelain slab **102**, described herein. In another example, the weight of porcelain slabs would conventionally fracture, or damage, by the process of stacking, or layering, another porcelain slab thereon as illustrated in FIG. **6A**. The weight of the stacked porcelain slabs would cause the lowest positioned porcelain slab to break, chip, or crack from the weight of the stack. Each of these stresses has challenged industrial usage of large size panels as the porcelain slab **102** is often

broken, chipped, or fractured, prior to installation. However, as illustrated, the structural core board **130** may advantageously deform to absorb, or evenly distribute, the weight of the stacked reinforced porcelain panel products **100** to reduce the breakage, chipping, and/or cracking of the porcelain slab **102**. By way of illustration, the structural core board **130** of each of the reinforced porcelain panel products **100** of FIG. 6A are numbered as **640**, **642**, **644**, and **646**. In this example, a visual representation of the deformation of each of the structural core boards **640**, **642**, **644**, **646** is shown. The stacked reinforced porcelain panel products **100** exert the compiled weight of reinforced porcelain panel products **100** above and thus the structural core board **640** is illustrated with the most deformation to absorb the weight of the stack. Similarly, structural core board **642** is illustrated with the second most deformation to absorb the weight of the stack, structural core board **644** is illustrated with the third most deformation to absorb the weight of the stack, and structural core board **646** exhibits the least deformation as no other reinforced porcelain panel product is positioned thereon. Stated differently, the thickness of the structural core boards **640**, **642**, **644**, **646** each are shown to indicate the weight absorption such as the structural core board **640** having a thickness less than the structural core board **642**, which has a thickness less than the structural core board **644**, which has a thickness less than the structural core board **646** to advantageously reduce the breakage, chipping, and/or cracking of the porcelain slab **102**.

(54) Similarly, as shown in FIG. 6D, operations at fabrication or distribution sites **620** also have hazards that may break, chip, or fracture the reinforced porcelain panel products **100**, or **400**. For example, a pallet of reinforced porcelain panel products **600** may be separated for fabrication and distribution to installation sites. Workers **622** may utilize other tools to lift each of the reinforced porcelain panel products **100**. For example, a tool may be a rail **624** and suction cup **626** tool to hoist a top reinforced porcelain panel product **100** off the pallet of reinforced porcelain panel products **600** and move to a desired location. Operations at fabrication or distribution sites **620** may also decorate or manipulate the reinforced porcelain panel products to cut and/or display an aesthetic feature, as discussed above. Furthermore, the workers **622** may load the reinforced porcelain panel product **100**, **400** onto a delivery truck **628** with a transport device, such as a crate **630**, further including an a-frame crate embodiment, positioned thereon to deliver the reinforced porcelain panel products **100**, **400** to the installation site. In one embodiment, the reinforced porcelain panel products **100**, **400** may be stacked at an angle of the crate **630** to encourage inward leaning and strapped to the crate with straps **632**. However, conventional unsupported porcelain slabs frequently experienced integrity failure as the leaning and strapped positioning stressed the fragile stone leading to breaks, chips, and/or fractures. For example, while not illustrated, the leaning and strapped positioning may experience 1) a force from the ground by the weight of the porcelain slab and may further experience 2) a force from the weight of another porcelain slab leaning on the first porcelain slab. To rectify the problem of integrity failure, mentioned above, the reinforced porcelain panel products **100** possess the structural core board **130** that may deform by 1) compressing the periphery of the structural core board **130** to absorb the weight of the reinforced porcelain panel product **100** and may further deform by 2) compressing the structural core board **130** of the stacked, or layered, second reinforced porcelain panel product **100** leaning thereon to advantageously reduce the breakage, chipping, and/or cracking of the porcelain slab **102**.

(55) Each of the movements of the reinforced porcelain panel products **100** is an added risk to mishandling leading to loss of product. Hazards during operations at fabrication or distribution sites **620** may include dropping of the reinforced porcelain panel product **100**, **400** onto a hard surface, colliding the reinforced porcelain panel product **100**, **400** with a hard surface, or bending the reinforced porcelain panel product **100**, **400** from improper use of the rail **624** and suction cup **626** tool. Similarly, each of these stresses on the porcelain slab **102** has previously challenged industrial usage of large size panels as the porcelain slab **102** is often broken, chipped, or fractured, prior to installation. However, the structural core board **130** integrated onto the porcelain slab **102** has provided a layer of protection against these likely stresses from handling during installation

reducing product loss and construction cost by for example, enabling the structural core board **130** to deform to absorb the contact or weight of stacked, or layered reinforced porcelain panel products **100** that advantageously reduce the breakage, chipping, and/or cracking of the porcelain slab **102**. (56) FIG. 7A-7E are exemplary cutting operations **700** of the reinforced porcelain panel product **100**, according to one embodiment of the present disclosure. FIG. 7A-7C illustrate a scoreline technique and FIGS. 7D and 7E illustrate edge cutting techniques.

(57) FIG. 7A illustrates a stone mason **702** with a scoring tool **704** in hand. The stone mason **702** is creating a scoreline **706** on the top surface **104** of the porcelain slab **102** within the reinforced porcelain panel product **100**, or **400**. The scoring tool may be a glass cutting wheel, a porcelain or tile cutter, a scoring wheel, or the like, as will be understood by those skilled in the art, commonly used to score ceramics, stone, and/or glass. After the stone mason **702** has positioned the scoreline **706** in the desired location, the stone mason **702** may initiate an angular bend **710** across the reinforced porcelain panel product **100**, as shown in FIG. 7B, or a radial force exerted on or near the scoreline **706**, such that the edges **708** move towards each other in a direction facing opposite the opening of the scoreline **706** causing a snap **720** that controllably breaks the porcelain slab **102** into two portions, the first portion **102a** and the second portion **102b**. In one embodiment, the snap **720** separates the porcelain slab **102** into two portions however the structural core board **130** remains intact. In one embodiment, the structural core board **130** may be cut with a utility knife **712**, or other similar cutting tool to separate the reinforced porcelain panel product **100** into two separated portions (not shown). In another embodiment, the snap **720** separates the porcelain slab **102** and the structural core board **130** into two portions. In some embodiments, a score to cut technique may cut the reinforced porcelain panel product **100** in less than about a minute, such as about 30-45 seconds, as compared to conventional porcelain cutting times of 30 minutes or more.

(58) The cutting of the reinforced porcelain panel product **100**, as described, is advantageous as the speed of cutting a porcelain slab is significantly increased. The reinforced porcelain panel product **100** may also be cut by use of conventional cutting tools, such as a bridge-saw using a diamond blade, a water jet, an edging machine used to grind panel edges, a CNC router, or the like, as would be understood by one skilled in the art. Conventionally, cutting techniques using saws for porcelain slabs **102**, for example, necessitate the tuning of the cutting speed and water cooling to reduce the damage to porcelain slabs. For example, a stone mason **702** may have to adjust the cutting wheel and the cooling water flow of the cutting tool to a slow speed/high flow to reduce the chipping effect resulting from high-speed, or high heat, of cutting porcelain. In comparison, the stone mason **702** may not be required to slow the cutting speed of the cutting tool, and/or adjust the cooling water flow, for other stone materials, such as a quartz or granite panel. Based on results of experimentation of the disclosed reinforced porcelain panel product **100**, it was found that the reinforced porcelain panel product **100** may be cut about four to five times faster than a quartz or granite panel in manner described above. In one embodiment, a fabricator may cut the reinforced porcelain panel product **100** using a diamond bladed bridge-saw at a speed of up to 30 feet per minute ("fpm"), such as about 3 fpm to about 30 fpm, such as about 15 fpm to about 27 fpm, such as about 20 fpm to about 25 fpm, thereby improving upon traditional porcelain cutting times. Therefore, the added stability from the structural core board **130**, or interlocking backing segments **430a-c**, enables a stone mason **702** to portion, or cut, the reinforced porcelain panel product **100** with a cutting technique discussed above, thereby to significantly reduce cutting times and save overall water usage.

(59) FIGS. 7D and 7E are exemplary edge cutting techniques in which a stone mason **702** may utilize in the fabrication of the specifications or during installation. For example, FIG. 7D shows a utility knife **712** providing a bevel edge, as shown by line **722**. As discussed above, the structural core board **130** may be configured to be cut such that beveled edges may be produced with a cutting tool. For example, a 45 degree angle cut across the structural core board **130** may allow the reinforced porcelain panel product **100** to be installed on a wall, as will be discussed in FIGS. 8 and

9. In another example, the structural core board **130** may be cut back to provide porcelain slab contact between two porcelain slabs **102** of adjacent reinforced porcelain panel product **100**, thereby to install a contacting seam along the length, or width, of the adjacent reinforced porcelain panel products.

(60) In another embodiment, as illustrated in FIG. 7E, the reinforced porcelain panel product **100** may be beveled along the line **724** through both the porcelain slab **102** and structural core board **130**. In this embodiment, a suitable cutting tool, such as an angle grinder, a bridge-saw, or the like, may cut through the porcelain slab **102** and structural core board **130** to provide an angle for installation. For example, a corner detail, such as a trim piece discussed in FIG. **10**, or a kitchen counter, as illustrated in FIG. **11** may abut in a desirable position such that two porcelain slabs **102** of adjacent reinforced porcelain panel products **100** mate by contacting their beveled edges along their respective lines **724**. While the illustrations of FIGS. 7D and 7E show beveled edges, the cuts may be performed using other shapes, such as, but not limited to, a chamfer, square, bullnose, knife cut, roman ogee, rounded, mitered, eased, cove, or the like, or a combination thereof for intricate or detailed installations that showcase the porcelain slab **102** of the reinforced porcelain panel products **100** as desired.

(61) FIG. **8** is an exemplary installation **800** of the reinforced porcelain panel product **100**, according to one embodiment of the present disclosure. FIG. **8** illustrates a worker **812** leaning a reinforced porcelain panel product **100** into position against the wall **802** in a direction **810**. In one embodiment, the wall **802** may be a studded wall, for example, a framed wall using studs. In one embodiment, as shown in FIG. **8**, the wall **802** may have a gypsum panel **804**, or drywall, positioned between an installed reinforced porcelain panel product **100** and the wall **802**. In one embodiment, the reinforced porcelain panel product **100** may be a gypsum panel **804** substitute and thus a gypsum panel **804** is not necessary for installation. The reinforced porcelain panel product **100** advantageously does not require the usage of thinset mortar material, but rather uses construction adhesive to connect the wall **802** to the reinforced porcelain panel product **100**. As will be further discussed below in FIG. **12**, as the structural core board **130** is a panel not governed by a national tile counsel and therefore, the reinforced porcelain panel product **100** may be glued to the intended surface with dollops **820** of construction adhesive spaced about 8 inches to about 16 inches apart. Prior to installation, the worker **812** may skillfully remove a portion **808** from the porcelain slab **102** and the structural core board **130** to create a void space configured to fit a fixture **806**, such as a plumbing tube therein, causing the fixture to protrude out from of the top surface **104** of the porcelain slab **102**. In one embodiment, as would be understood by one skilled in the art, the fixture **806** may be a pipe configured to receive a finished fixture thereon after installation of the reinforced porcelain panel product **100** around or about the pipe. Furthermore, the worker **812** may trim, or remove, any excess structural core board **130** overhanging the desired dimensions of the porcelain slab **102** with a utility knife **712**, or other similar cutting tool. In one embodiment, the structural core board **130** overhang is trimmed to abut against another structural core board **130** of a second reinforced porcelain panel product **100**, thereby to define an intended grouting joint thickness while protecting both reinforced porcelain panel products **100** from colliding. In another embodiment, the reinforced porcelain panel product **100** may be advantageously shimmed into a level installation position using shims, as would be understood by one skilled in the art, thereby to enable an increased speed of installation. FIG. **8** installation **800** may provide for usage of thin large format porcelain for various installations, such as bathrooms (as shown in FIG. 9A-D), kitchens (as shown in FIG. **11**), and/or commercial spaces, such as a hotel lobby (as shown in FIG. **12**).

(62) FIGS. 9A-D are exemplary installations of the reinforced porcelain panel product in a bathroom setting, according to various embodiments of the present disclosure. FIG. 9A illustrates a worker **912** leaning a porcelain slab **102** to be installed onto the stud wall **902** having bathroom fixtures positioned thereon. As shown, the bathroom **900** includes a pair of stud walls **902**, a

gypsum paneled wall **904**, bench **922** (unfinished), a floor **920**, and bathroom fixtures. The bathroom fixtures include a shower head **914** and faucets **916**. It is to be understood a variety of fixtures may be installed in a bathroom **900** which require a portion **908** removed from the reinforced porcelain panel product **100** to fit the reinforced porcelain panel product **100** onto the stud wall **902**, for example. In one embodiment, the reinforced porcelain panel product **100** may be directly attached to the wall **902**, or the wall **904**, with adhesive. In one embodiment, the reinforced porcelain panel product **100** may abut a water proofing barrier (not shown), such as Redguard® or waterstop as would be understood by one skilled in the art, positioned on a floor **920** and between the reinforced porcelain panel product **100**, the stud wall **902**, and the gypsum paneled wall **904** when installed. In one embodiment, the structural core board **130** of the reinforced porcelain panel product **100** may be coated with a waterproof coating to create a watertight enclosure when the reinforced porcelain panel product **100** is installed onto the wall **902**, or wall **904**. In one embodiment, the waterproof coating may be rolled, sprayed, or placed on the structural core board **130** with a large format roll of tape, or the like, as will be understood by those skilled in the art. In yet another embodiment, the waterproof coating may be rolled, sprayed, or placed on the wall **902** such that the reinforced porcelain panel product **100** is installed onto the wall **902** with the waterproof coating.

(63) Generally, large panels of material are desired in bathroom settings as more wall coverage is available with the large panels. Therefore, thin large format porcelain is desirable for such applications due to a reduction of necessary grout, or caulking, compared to conventional tiling of the same walls and also due to the ability of porcelain to withstand wear and tear of showers when installed. For example, a finished shower room **901** using the thin large format porcelain, as shown in FIG. **9B**, may use grout, or caulking, at the interfacing joints **918**. The reduction of use of grout, or caulking, is advantageous as both, grout and caulking are prone to mold, crack, and/or fail causing water damage behind the stone, tile, or panel installed. The use of the reinforced porcelain panel product **100** may span the coverage each of the wall **902**, the wall **904**, the floor **920**, the bench **922** (finished), and a shelf **924** (as shown in FIG. **9B**) with a reduced usage of grout, or caulking, at the interfacing joints **918**.

(64) Previously, thin large format porcelain was limited to smaller panel sizes and unable to provide full coverage for shower walls as the thin large format porcelain was fragile to handle in large sizes. However, the implementation of the structural core board **130** onto the porcelain slab **102** has enabled the size of the reinforced porcelain panel product **100** to increase to provide larger panels previously unattainable by porcelain slabs **102** alone. Therefore, installation of the reinforced porcelain panel product **100**, or **400**, may be easier and quicker to install as larger panels provide greater coverage as compared to prior porcelain slabs, alternative stone, or tile solutions. For example, an installation may be increased by a reduction of preparation time, such as squaring off walls, and shimming the reinforced porcelain panel product **100** thereby to stabilize the material resulting in less damage to the porcelain, as would be understood by one skilled in the art. Furthermore, fewer materials may be required for installation further increasing the ease and speed of installation. For example, in one embodiment, the wall **902** may be finished with only the reinforced porcelain panel product **100** compared to the conventional method of installation that requires tiles attached, via thinset mortar, to a gypsum panel that is fastened to the wall and a waterproof barrier, or coating, positioned between the tile and the gypsum panel. Thus, the speed of installation of the reinforced porcelain panel product **100**, as discussed, increases project completion efficiency, and also provides the owner less maintenance as less grout is used. In one example, the installation of a shower room **901** may be performed in less than about 24 hours, such as less than about 20 hours, such as less than about 15 hours, such as less than about 10 hours, such as less than about 5 hours, such as less than about 3 hours, such as about 3 hours.

(65) FIG. **9C** illustrates one panel assembly embodiment for magnetic shower attachments through the reinforced porcelain panel product **100**, according to one embodiment of the present disclosure.

As discussed above, shower rooms **901** and fixtures may be adorned with a reinforced porcelain panel products **100**, however, the use of grout, or caulking, may be necessary at the interfacing joints **918**. Furthermore, conventional shower accessories may necessitate drilling into the porcelain slab **102** to secure the accessory to the wall **902**, thereby to create a water infiltration site needing to be caulked to prevent water damage. One embodiment of the present disclosure takes advantage of the thin porcelain slabs **102** that allow attractive magnetic forces to position movable magnetic accessories on the top surface **104** of the porcelain slab **102**, for example. For example, FIG. **9C** illustrates a partially exploded view of wall **902** having a metal panel **960** positioned behind the porcelain slab **102** when installed. It is to be understood, the metal panel **960** may further be positioned between the porcelain slab **102** and the wall **904** when installed. In one embodiment, the metal panel **960** may be replaced with a magnet such as a permanent magnet. In one embodiment, the metal panel **960** may be a magnetic material, such as steel, or other magnetic material. In those embodiments, magnetic forces may attract magnetic shower accessories **962** through the porcelain slab **102**. In one embodiment, the magnetic shower accessories **962** possess a magnet configured to be attracted to the metal panel **960**. In another embodiment, the magnetic shower accessories **962** possess a magnetic material to attract to a magnet positioned behind the porcelain slab **102**. In one embodiment, the metal panel **960** is a small panel positioned and installed in desired areas behind the installed reinforced porcelain panel product **100** and adhesively connected to the wall **902**. In one embodiment, the metal panel **960** is thin and does not add a significant difference in thickness to the wall **902** during construction. It is also contemplated that a non-metallic sheet, equal in thickness to the metal panel **960**, may be installed around the metal panel **960** such that the wall **902** is level. Fasteners, such as screws, clips, or the like, as will be understood by those skilled in the art, and adhesives, as discussed above, are contemplated methods of securing the metal panel **960** to the wall **902**. In one embodiment, the metal panel **960** is large such that a significant portion of the wall **902** is magnetic, thereby to provide a large surface area to adjust the magnetic shower accessories **962** about the wall **902** and into a desired position to fit an individual's need, which may be due to height, weight, temporary disability, permanent disability, aesthetic preference, or other functional preference. The magnetic shower accessory **962** may be a shelf, a shower caddy, a footrest, a towel rack, a shower arm for a handheld shower hose, or the like shower accessories, as will be understood by those skilled in the art. In one embodiment, the porcelain slab **102**, or the reinforced porcelain panel product **100**, are attached to the metal panel **960** with adhesive, such as construction grade adhesive.

(66) The magnetic shower attachments advantageously hide the metal panel **960** behind the porcelain slab **102** to discreetly provide functionality to the positioning of the magnetic shower accessories **962** while reducing the use of grout, or caulking, that challenges shower installations. While the magnetic shower attachments are illustrated connected and installed in a wall, other surfaces walls are contemplated, such as a ceiling or the floor **920**. Therefore, it may be understood an assembly for positioning a movable magnetic accessory may include a magnetic panel affixed to a mounting surface and the reinforced porcelain panel product positioned to conceal the magnetic panel and having thickness less than the magnetic attractive force configured between a magnetic accessory attachment positioned on a top surface of the porcelain slab and the magnetic panel.

(67) It is to be understood that the disclosure of the magnetic shower attachments as discussed in FIG. **9C** may also utilize other materials such as granite, quartz, ceramic, concrete, limestone, vinyl, marble, laminate, glass, fiberglass, or the like, as will be understood by those skilled in the art, rather than the porcelain slab **102**, or reinforced porcelain panel product **100**, to finish the wall **902** as a cover panel configured to conceal the magnetic metal panel, for example. In another embodiment, the cover panel may conceal the magnet, such as a permanent magnet installed behind the cover panel for use with magnetic material-containing shower attachments. In some embodiments, the other materials, mentioned above, may or may not possess an equivalent structural core board, such as structural core board **130** used in the reinforced porcelain panel

product **100**. Furthermore, while the above disclosure of the magnetic attachments was made in reference to a shower embodiment, it is contemplated the discrete metal panel **960** may be used in other spaces such as, for example, shelving in water closets, on vanity walls, in living rooms, above fireplaces, and/or the like.

(68) FIG. **9D** illustrates one embodiment that uses reinforced porcelain panel product **100** as wall cladding **944** and a wainscoting **942** on wall **940**. Wall cladding **944** may be panels of reinforced porcelain panel product **100** installed on the wall **940** that was previously, for example, tiled or finished, with only adhesive. In some embodiments, fasteners may also be used to attach the wall cladding **944**. However, as briefly discussed above and further discussed below, the structural core board **130** is a panel not governed by a national tile counsel and therefore, the reinforced porcelain panel product **100** may be glued to the intended surface using only dollops **820** of construction adhesive, as shown in FIG. **8**. In one embodiment, the wall cladding **944** is decorated to display an appealing design to add visual appeal to the bathroom **900**, for example. Similarly, the wainscoting **942** may be designed with several aesthetic, or artistic, designs to appeal to the desires of the owner.

(69) FIG. **10** is an exemplary installation of a trim piece **1040** constructed from a reinforced porcelain panel product **100**, according to one embodiment of the present disclosure. The reinforced porcelain panel product **100** may be cut into narrow sections **1030** and adhesively attached on one end **1032** of each of the narrow sections **1030** of the reinforced porcelain panel product **100**, thereby to define the trim piece **1040**. In one embodiment, the trim piece **1040** may be configured to fit around a corner **1034** to provide a decorated corner **1034** using the trim piece **1040**. In one embodiment, the trim piece **1040** may extend a distance of about 1 inch in each direction covering the corner **1034**. In one embodiment, the trim piece **1040** may have the shape of a sharp corner or a rounded corner, although various shapes are contemplated. In one embodiment, the trim piece **1040** may abut to a reinforced porcelain panel product **100** installed on the wall having the corner **1034**. As discussed above, porcelain may be manipulated to display a variety of patterns enabling an owner to create a unique design that appeals to the owner. Here, the trim piece **1040** may match the wall decoration or may provide a varied pattern for an accented piece on the corner **1034**. Furthermore, in one embodiment, the trim piece **1040** may have the structural core board **130** removed for installation such that no overhang is present when installing the trim piece **1040**.

(70) FIG. **11** is an exemplary installation of a countertop **1108** constructed from a reinforced porcelain panel product **100** in a kitchen **1100** with a cutaway **1114** showing the structural core board **130** of the installed countertop **1108**, according to one embodiment of the present disclosure. As illustrated, the worker **912** may position reinforced porcelain panel product **100** onto a countertop support **1112** to construct the countertop **1108**. In one embodiment, the countertop **1108** may have a cutout **1110** to accommodate a kitchen sink **1102** and may further have other cutouts to position a faucet spout **1104** and faucet handles **1106** near the kitchen sink **1102**. The countertop **1108** may be secured with fasteners or adhesive to the countertop support **1112**. Furthermore, as illustrated, the countertop **1108** may be finished with a trim **1116**, for example, a trim piece **1040** further customized for an edge of a countertop **1108**. As shown, the trim **1116** may be a portion of a reinforced porcelain panel product **100** cut at, for example, a 45-degree angle and to mate with a 45-degree angle cut on the countertop **1108**. The two 45-degree angles may be joined by porcelain-compliant adhesive, or the like, as would be understood by one skilled in the art. Similarly, the trim **1116** may be joined with the adjacent trim **1116** such that both trims come to an edge that further aligns with the countertop **1108** to construct a point **1118**. While the above cuts are referenced to have 45-degree angles, the cuts may be made with various angles common in the countertop industry. The structural core board **130** advantageously supports both the adjacent structural core board **130** and the porcelain slab **102** of both the countertop **1108** and the trim **1116**, thereby providing an added strength to the joint between the two trim **1116** and the countertop **1108**, such

as at the point **1118**.

(71) FIG. **12** is an exemplary installation of a tiled art piece **1202** displayed on a display wall **1204** constructed from a reinforced porcelain panel product **100**, according to one embodiment of the present disclosure. As discussed above, a pattern may be designed on an intact reinforced porcelain panel product **100** at a fabrication location. As illustrated, the pattern **1208** is represented as a natural stone vein pattern across the top surface **104** of the porcelain slab **102** for display on the display wall **1204**. A stone mason (not shown) may utilize a cutting operation, as discussed in FIG. **7A-C**, to create individual smaller panels, or tiles **1210** divided by tile lines **1206**, that may allow for easier shipping and easier installation as compared to installing a large intact reinforced porcelain panel product **100**. In one embodiment, the tiles **1210** may be labeled with an identifier, such as an alpha or numeric identifier, which allows the tiles **1210** to be placed in a particular order when installed to create the originally intended layout or aesthetic design created at, for example, a fabrication location. Furthermore, as discussed above, the reinforced porcelain panel product **100** may weigh significantly less than other stone or ceramic alternatives and therefore may cause less stress, and/or damage to mounting surfaces, such as a display wall **1204**. For additional perspective, the tiled art piece **1202** may be placed in a commercial space, such as a hotel lobby.

(72) Traditionally, conventional tiles (not shown) are installed onto cement, concrete, or the like hard surfaces, as will be understood by those skilled in the art, with an adhesive mortar, such as a thinset mortar. Thinset mortar, commonly referred to as thinset cement, dryset mortar, or drybond mortar by those skilled in the art, is an adhesive mortar made of cement, fine sand, and a water-retaining agent such as an alkyl derivative of cellulose to bond two hard surfaces together, such as a ceramic tile to a hard surfaced wall. The installation of conventional tiles (not shown) is directed by policies set forth by the Tile Council of North America (TCNA), a trade organization, which requires the use of thinset mortar for installation of tile, including conventional porcelain. The use of thinset mortar may be problematic as voids, or cavities, between the installed material and the wall may be prone to breaking as the void is unsupported by the thinset mortar leading to future repairs. However, the use of the reinforced porcelain panel product **100** may not require the use of thinset mortar as the structural core board **130** is the adhesive surface, rather than the porcelain slab **102** which required thinset mortar. Therefore, as the reinforced porcelain panel products **100** are lighter than other stone alternatives and cause less stress, and/or damage, to the mounting wall, an adhesive, such as a construction grade adhesive, may be used to secure the tiled art piece **1202** onto the display wall **1204**, thereby to reduce the amount of clean up from thinset derived dust or splatter, offsetting any additional costs of materials, such as the cost of thinset mortar and adhesive, and increase the speed of installation to thereby increase project efficiency.

(73) FIGS. **13A** and **13B** contain a flow chart of a method **1300** of fabricating a reinforced porcelain panel product **100** for commercial or residential renovations or new constructions, according to one embodiment of the present disclosure. Method **1300** includes blocks **1302**, **1304**, **1306**, **1308**, **1310**, **1312**, **1314**, **1316**, and **1318**. While the operations are illustrated sequentially, some of the operations may be excluded to achieve owner designs. As discussed above, the embodiments of the reinforced porcelain panel product **100** equally apply to the method discussed below.

(74) Method **1300** begins with optional block **1302** by, for example, interlocking two or more interlocking backing segments to construct a structural core board. As discussed above in FIG. **4**, the structural core board **130** may be constructed to a single structural core board by use of various interlocking backing segments that utilize, for example an overlay joint, such as a finger, shiplap, or a dovetail joint, among other types of overlay joints, configured to join or fasten the interlocking backing segments **430a-430c** together.

(75) Method **1300** continues to block **1304** by applying an adhesive material **250** to foam-core backing material, such as structural core board **130**. The adhesive material **250** may be applied by use of an aerosol adhesive tool, a roller, an applicator, and spread using a spreading tool, or the

like, as would be understood by one skilled in the art.

(76) At block **1306**, a worker may continue method **1300** by abuttingly contacting the structural core board **130** to a bottom surface **106** of a porcelain slab **102** with an adhesive material **250** therebetween such that regions of the top surface **134** of the structural core board **130** substantially extend beyond outer peripheries of a first surface area **108** of the porcelain slab **102** to define buffer regions **260** so that a combination of the porcelain slab **102** and the structural core board **130** define the reinforced porcelain panel product **100**, thereby to allow the buffer regions **260** to be visible when viewing the reinforced porcelain panel product **100** from a front view of a top surface **104** of the reinforced porcelain panel product **100**, as shown in FIG. 2B, and to enhance protection from cracking when the reinforced porcelain panel product is handled. Similarly, at block **1306**, a worker may utilize another embodiment of the reinforced porcelain panel product **100**, such as illustrated in FIGS. 2C, 4B, 6C, 7D and 7E, wherein the structural core board **130** and the porcelain slab **102** are flush aligned. Stated differently, a worker, at block **1306**, may continue method **1300** by abuttingly contacting the structural core board **130** to a bottom surface **106** of a porcelain slab **102** with an adhesive material **250** therebetween such that regions of the upper surface **134** of the structural core board **130** substantially align with outer peripheries of a first surface area **108** of the porcelain slab **102**, thereby to enhance protection from cracking when the reinforced porcelain panel product is handled. In one embodiment, the bottom surface **136** of the structural core board **130** has a first surface plane substantially parallel to the top surface **104** of the porcelain slab **102** such that the reinforced porcelain panel product **100** is rectangular. In one embodiment, the bottom surface **136** of the structural core board **130** is configured to adhesively attach to a mounting surface such as a wall, as shown in FIGS. 8-12. In one embodiment, the structural core board **130** has a second surface area **138** greater than the first surface area **108** of the porcelain slab **102** and has a thickness **240** greater than the porcelain slab **102**. As discussed above, the porcelain slab **102** integrated within the reinforced porcelain panel product **100** may be manipulated to incorporate a variety of designs, patterns, or the like, as will be understood by those skilled in the art, to create an aesthetic décor for the owner of the commercial, or residential, space, such as adhesively attaching trims to construct a wainscoting, as discussed above.

(77) Method **1300** may continue at optional block **1308** where the bottom surface **136** of the reinforced porcelain panel product **100** may be coated such that the bottom surface **136** is substantially non-permeable to air and thereby to allow the use of a vacuum lifting tool to move the reinforced porcelain panel product **100**. In one embodiment, the structural core board **130** has a coating that is substantially waterproof. The waterproof coating is advantageous in embodiments that are installed in wet areas, such as shower enclosure floor where waterproofing is a desirable characteristic to reduce water damage from lack of water containment, as would be understood by one skilled in the art. The vacuum lifting tool may be an air-driven table, or “air table”, able to sufficiently lift the weight of the reinforced porcelain panel product **100** to aid a worker in the movement and/or positioning of the reinforced porcelain panel product **100**. In another embodiment, the vacuum lifting tool may a vacuum table that provides suction configured to restrain the reinforced porcelain panel product **100** from movement and/or positioning of the reinforced porcelain panel product **100** while the reinforced porcelain panel product **100** is being processed, such as decorated, scored, or cut.

(78) The method **1300** may pursue various pathways, such as pathways **1350**, **1352**, and/or **1354**. For example, pathway **1350** may be utilized for shower enclosures as discussed above in FIGS. 8 and 9A-9D. In another example, pathway **1352** may be utilized for a tile **1210** embodiment, or a trim piece **1040** embodiment, produced from the reinforced porcelain panel product **100** as discussed above in FIGS. 10 and 12. In yet another example, pathway **1354** may be utilized for a kitchen counter embodiment as discussed above in FIG. 11. Each of the pathways **1350**, **1352**, and/or **1354** may conclude the method **1300** at block **1318**.

(79) At block **1310**, a worker, or stone mason, may score the reinforced porcelain panel product

100 such as, in one embodiment, scoring the top surface **104** of the porcelain slab **102** with one or more scorelines **706** positioned along one or more selected locations across the top surface **104**. The scorelines **706** provide guided and controlled breaks to snap **720** the porcelain slab **102** into a selected pattern when a radial force is exerted on or near the scoreline **706**.

(80) At block **1312**, a worker, or stone mason, may cut the reinforced porcelain panel product **100** configured to fit kitchen fixtures therein such that the reinforced porcelain panel product **100** is laid on a kitchen counter support structure to be used as a kitchen counter as discussed in FIG. **11**. The worker, or stone mason may utilize cutting techniques described in FIGS. **7A-D** and/or may use tools to perform porcelain cuts, such as water jet, an angle grinder, or the like, as would be understood by one skilled in the art.

(81) At block **1314**, a worker, or stone mason, may cut the reinforced porcelain panel product **100** by using, for example, cutting techniques described in FIGS. **7A-D** to cut the reinforced porcelain panel product **100** into various sized or shaped tiles as discussed above in FIG. **12**.

(82) At block **1316**, the cut tiles from the reinforced porcelain panel product **100** may be positioned in a selected, or desired, orientation such that the reinforced porcelain panel product **100** may be reassembled in the selected orientation desired pattern at a second location for installation as discussed above in FIG. **12**. Stated differently, the cut tiles from the reinforced porcelain panel product **100** of block **1314** may be positioned in a selected orientation and may further be marked in such a way that enable a tile installer to reconstruct the intended design, i.e., the selected orientation.

(83) The various embodiments of method **1300** may conclude at block **1318** where the bottom surface **136** of, for example, the kitchen counter, the trim pieces or tiles, or the shower enclosure panels using reinforced porcelain panel product **100** containing an embodiment of the structural core board **130**, are adhesively mounted to the commercial or residential surface, such as display wall **1204** shown in FIG. **12**, to display the top surface **104** of the porcelain slab **102**. As discussed above, the bottom surface **136** of the structural core board **130**, and thus the reinforced porcelain panel product **100**, may be attached to the desired mounting surface with adhesive, such as construction grade adhesive.

(84) FIG. **14** is a flow chart of a method **1400** of transporting a reinforced porcelain panel product **100**, according to one embodiment of the present disclosure. Method **1400** includes blocks **1402**, **1404**, and **1406**. While the operations are illustrated sequentially, some of the operations may be excluded to achieve owner designs. As discussed above, the embodiments of the reinforced porcelain panel product **100** equally apply to the method discussed below.

(85) Method **1400** begins at block **1402** by positioning a reinforced porcelain panel product **100**, defined above, onto a transport device, such as, but not limited to, a pallet **602**, a freight shipping **610**, a crate **630**, a forklift, or an a-frame crate, as shown and discussed in FIGS. **6B** and **6D**.

(86) The method **1400** may continue at block **1404** with stacking two or more reinforced porcelain panel products **100** such that each of the structural core boards **130** deform to protect the porcelain slab **102** from chipping or cracking from a force exerted onto the reinforced porcelain panel product **100** by a ground or a weight of from another reinforced porcelain panel product **100** stacked thereon. As discussed in FIGS. **6A-6D**, the structural core boards **130** deforms to advantageously reduce the breakage, chipping, and/or cracking of the porcelain slab **102** as compared to conventional unsupported porcelain.

(87) The method **1400** continues to a remote location, such as a second location different than the first location, at block **1406**, by transporting the reinforced porcelain panel product **100** to a recipient at a remote location such that the recipient receives the reinforced porcelain panel product **100** void, or absent, of breaks on the porcelain slab **102**. In one embodiment, the fabricated reinforced porcelain panel product **100** may be in one or more reinforced porcelain panel product sections, such discussed in the cutting embodiment of the reinforced porcelain panel product **100** above.

(88) FIG. 15 is a flow chart of a method for lifting a reinforced porcelain panel product using a lifting clamp, according to one embodiment of the present disclosure. Method **1500** includes blocks **1502**, **1504**, **1506**, **1508**, and **1510**. While the operations are illustrated sequentially, some of the operations may be excluded to achieve owner designs. As discussed above, the embodiments of the reinforced porcelain panel product **100** equally apply to the method discussed below.

(89) Method **1500** begins at block **1502** by positioning a reinforced porcelain panel product **100**, discussed above, within an interstitial space **550** of the lifting clamp **502** configured to open or close the interstitial space **550** by actuation of the lifting clamp **502**. As discussed above in FIGS. **6A-6D**, the conventional fabrication shops may use heavy equipment, such as the lifting clamp **502**, to lift heavy panels that may not be sufficiently delicate to handle and carry thin panels, such as an unsupported porcelain slab. However, by use of the present disclosure of a reinforced porcelain panel product **100**, the deformation of the structural core board **130** provides protection against breakage, chipping, and/or cracking of the porcelain slab **102** when using a lifting clamp **502**, as discussed in FIGS. **5A** and **5B**.

(90) The method **1500** may continue at block **1504** with contacting the reinforced porcelain panel product **100** with the lifting clamp **502** such that the porcelain slab **102** and the structural core board **130** each abuttingly contact the movable jaws **552** of the lifting clamp **502**. The movable jaws **552** of the lifting clamp may be actuated to close the interstitial space **550**, thereby to contact the reinforced porcelain panel product **100** via the porcelain slab **102** and the structural core board **130**, respectively.

(91) The method **1500** continues at block **1506** by compressing the reinforced porcelain panel product **100** such that the structural core board **130** deforms to protect the porcelain slab **102** from chipping or cracking caused by the compression of the movable jaws **552** of the lifting clamp **502**. In one embodiment, the compression of the movable jaws **552** of the lifting clamp **502** may exert a force that would fracture or crush conventional unsupported porcelain. In examples where the conventional unsupported porcelain was able to withstand the compression of the lifting clamp **502**, the porcelain often broke upon a lifting of the lifting clamp **502** with the porcelain held within. However, the structural core board **130** of the present disclosure advantageously provides protection against breakage, chipping, and/or cracking of the porcelain slab **102** when using a lifting clamp **502**. As illustrated in FIG. **5B**, the deformation of the structural core board **130** provides 1) grip for the lifting clamp, and 2) an ability to withstand the compression of the movable jaws **552** of the lifting clamp **502** to enable the lift.

(92) The method **1500** continues at block **1508** by lifting the reinforced porcelain panel product using the lifting clamp to elevate the reinforced porcelain panel product **100** from, for example, the ground, or crate, as discussed in FIGS. **6A-6D**.

(93) The method **1500** may conclude at block **1510** by moving the reinforced porcelain panel product **100** to a location different than the location of the lifting of the reinforced porcelain panel product **100** without breakage, chipping, and/or cracking of the porcelain slab **102**.

(94) FIGS. **16A** and **16B** are perspective views of a reinforced porcelain panel product kit, according to two embodiments of the disclosure. In all embodiments, the reinforced porcelain panel product kit **1600** includes a container **1630**, or a support member. The container **1630** may be a wooden or plastic box, a crate, a pallet, an a-frame crate, or the like, wherein the container **1630** is configured to house, support, or keep the components within the reinforced porcelain panel product kit **1600** together or in a specific position, to define a shipping package, as would be understood by those skilled in the art. In one embodiment, the container **1630** has a locking feature or a tamper-resistant feature to reduce unauthorized or inadvertent opening of the container **1630**. In another embodiment, the container **1630** has straps **1632** configured to position or restrain the contents of the reinforced porcelain panel product kit **1600** from moving for transport.

(95) FIG. **16A** illustrates an embodiment of a reinforced porcelain panel product kit **1600** for a shower enclosure that includes three porcelain slabs **1602**, **1604**, and **1606** positioned within the

container **1630**. As illustrated, the three porcelain slabs **1602**, **1604**, and **1606** are positioned in the boundaries of a shipping package. The three porcelain slabs **1602**, **1604**, and **1606** are panels specifically sized to accommodate the three most common shower enclosure sizes, as would be understood by one skilled in the art. In some embodiments, the reinforced porcelain panel product kit **1600** includes two porcelain slabs of substantially similar size and one porcelain slab that is larger than the two smaller porcelain slabs. In other embodiments, the reinforced porcelain panel product kit **1600** includes two porcelain slabs of substantially similar length and one porcelain slab that has a length longer than the two porcelain slabs. In yet another embodiment, the reinforced porcelain panel product kit **1600** includes two porcelain slabs of substantially similar width and one porcelain slab that has a width longer than the two porcelain slabs. For example, in one embodiment, the reinforced porcelain panel product kit **1600** includes three porcelain slabs **1602**, **1604**, and **1606**, wherein 1) the porcelain slab **1602** is about 60 inches by about 74 inches, and 2) the porcelain slabs **1604** and **1606** are each about 31 inches by about 74 inches. In another embodiment, the reinforced porcelain panel product kit **1600** includes three porcelain slabs **1602**, **1604**, and **1606**, wherein 1) the porcelain slab **1602** is about 60 inches by about 96 inches, and 2) the porcelain slabs **1604** and **1606** are each about 31 inches by about 96 inches. In still another embodiment, the reinforced porcelain panel product kit **1600** includes three porcelain slabs **1602**, **1604**, and **1606**, wherein 1) the porcelain slab **1602** is about 60 inches by about 106 inches, and 2) the porcelain slabs **1604** and **1606** are each about 31 inches by about 106 inches.

(96) FIG. **16B** illustrates an embodiment of the reinforced porcelain panel product kit **1600** wherein the kit is provided by a support member, such as a pallet. In one embodiment, the reinforced porcelain panel product kit **1600** is provided in a plurality of pallets, as illustrated. In yet another embodiment, the reinforced porcelain panel product kit **1600** is provided on single pallet. In still another embodiment, each porcelain slab may be provided on individual pallets, thereby to delicately transport fragile porcelain slabs.

(97) FIG. **16C** illustrates an embodiment of the reinforced porcelain panel product kit **1600** wherein the kit is provided within a container. As previously mentioned, the container **1630**, or a support member. The container **1630** may be a box, a crate, a pallet, an a-frame crate, or the like, wherein the container **1630** is configured to house, support, or keep the components within the reinforced porcelain panel product kit **1600** together or in a specific position, to define a shipping package, as would be understood by those skilled in the art. In one embodiment, the reinforced porcelain panel product kit **1600** includes more components as illustrated in FIG. **16C** such as, for example, additional trim pieces for finishing work, as discussed in FIG. **10**. In another embodiment, the reinforced porcelain panel product kit **1600** includes fewer components as illustrated in FIGS. **16A**, **16B**, and **16C**. In still another embodiment, the reinforced porcelain panel product kit **1600** further includes a magnetic metal panel **960** as discussed in FIG. **9C**. Further, it is to be understood by those skilled in the art that the reinforced porcelain panel product kit **1600** also may include instructional manuals, video tutorials, adhesive, cleaning wipes or cloth, shims, and/or the like, configured to provide instruction, and/or tools, such as a utility knife or a cutting tool, to enable a stone mason to install the reinforced porcelain panel product **100** in a desired location. The components as illustrated within the reinforced porcelain panel product kit **1600** of FIGS. **16A**, **16B**, and **16C** are not representative of the actual size, or quantity, but rather are exemplary to view an embodiment of a kit.

EXAMPLES

(98) The following examples reveal the durability of the reinforced porcelain panel product **100** discussed above as will be understood by those skilled in the art. The reinforced porcelain panel product **100** was experimentally evaluated to exhibit strength and resilience as compared to conventional porcelain option available for construction purchase. Below are the findings of tested examples for breaking strength, impact resistance, and adhesive resilience as will be understood by those skilled in the art.

Example 1

(99) In a first example, a breaking strength of the reinforced porcelain panel product **100** was experimentally determined under International Organization for Standardization (“ISO”) test number 10545-4, as will be understood by those skilled in the art. The breaking strength of the reinforced porcelain panel product **100** was determined to yield increased strength with a conventional 6-millimeter neolith porcelain. The below table illustrates the specific ISO test findings.

(100) TABLE-US-00001 TABLE 1 Breaking strength increase using a structural core board from Example 1. NEOLITH PRO- PROP- PROD- AVERAGE Porcelain STRENGTH CEDURE ERTY UCT RESULTS 6-mm INCREASE ISO 10545-4 Breaking T92.130 1,960 700 180% Strength Nmm/mm Nmm/mm

(101) Table 1 illustrates the breaking strength increase of about 180% by using a structural core board, such as structural core board **130** discussed above on a conventional 6-millimeter neolith porcelain slab as compared to the conventional 6-millimeter neolith porcelain slab without the structural core board. Therefore, the results support a finding of at least 180% breaking strength increase, such as at least about 175% breaking strength increase, such as at least about 170% breaking strength increase, such as at least about 160% breaking strength increase, such as at least about 150% breaking strength increase, such as at least about 140% breaking strength increase, such as at least about 130% breaking strength increase, such as at least about 120% breaking strength increase, such as at least about 110% breaking strength increase, such as at least about 100% breaking strength increase, such as at least about 90% breaking strength increase, such as at least about 80% breaking strength increase, such as at least about 70% breaking strength increase, such as at least about 60% breaking strength increase, such as at least about 50% breaking strength increase, such as at least about 40% breaking strength increase, such as at least about 30% breaking strength increase, such as at least about 20% breaking strength increase, such as at least about 10% breaking strength increase, by using a structural core board, such as structural core board **130** discussed above on a conventional 6-millimeter neolith porcelain slab as compared to the conventional 6-millimeter neolith porcelain slab without the structural core board.

Example 2

(102) In a second example, an impact resistance of the reinforced porcelain panel product **100** was experimentally determined under European Standard (“EN”) test number 14617-9, as will be understood by those skilled in the art. The impact resistance of the reinforced porcelain panel product **100** was determined to yield increased resistance with a conventional 6-millimeter florim porcelain. The below table illustrates the specific EN test findings.

(103) TABLE-US-00002 TABLE 2 Impact resistance increase using a structural core board from Example 2. FLORIM RESIST- PRO- PROP- PROD- AVERAGE Porcelain ANCE CEDURE ERTY UCT RESULTS 6-mm INCREASE EN 14617-9 Impact T92.130 4.76 J 3.16 J 50.63% Resistance (Fracture Work)

(104) Table 2 illustrates the impact resistance increase of at least about 50% by using a structural core board, such as structural core board **130** discussed above on a conventional 6-millimeter florim porcelain slab as compared to the conventional 6-millimeter florim porcelain slab without the structural core board. Therefore, the results support a finding of at least 50% impact resistance increase, such as at least about 45% impact resistance increase, such as at least about 40% impact resistance increase, such as at least about 35% impact resistance increase, such as at least about 30% impact resistance increase, such as at least about 25% impact resistance increase, such as at least about 20% impact resistance increase, such as at least about 15% impact resistance increase, such as at least about 10% impact resistance increase, such as at least about 5% impact resistance increase, by using a structural core board, such as structural core board **130** discussed above on a conventional 6-millimeter florim porcelain slab as compared to the conventional 6-millimeter florim porcelain slab without the structural core board.

Example 3

(105) In a third example, an impact resistance of the reinforced porcelain panel product **100** was experimentally determined under European Standard (“EN”) test number 14617-9. The impact resistance of the reinforced porcelain panel product **100** was determined to yield increased resistance with a conventional 12-millimeter florim porcelain. The below table illustrates the specific EN test findings. Unless otherwise notated, the unit “J” refers to Joules, as would be understood by one skilled in the art.

(106) TABLE-US-00003 TABLE 3 Impact resistance increase using a structural core board from Example 3. FLORIM RESIST- PRO- PROP- PROD- AVERAGE Porcelain ANCE CEDURE ERTY UCT RESULTS 12-mm INCREASE EN 14617-9 Impact T92.130 4.76 J 3.03 J 57.10% Resistance (Fracture Work)

(107) Table 3 illustrates the impact resistance increase of at least about 57% by using a structural core board, such as structural core board **130** discussed above on a conventional 12-millimeter florim porcelain slab as compared to the conventional 12-millimeter florim porcelain slab without the structural core board. Therefore, the results support a finding of at least 57% impact resistance increase, such as at least about 55% impact resistance increase, such as at least about 45% impact resistance increase, such as at least about 40% impact resistance increase, such as at least about 35% impact resistance increase, such as at least about 30% impact resistance increase, such as at least about 25% impact resistance increase, such as at least about 20% impact resistance increase, such as at least about 15% impact resistance increase, such as at least about 10% impact resistance increase, such as at least about 5% impact resistance increase, by using a structural core board, such as structural core board **130** discussed above on a conventional 12-millimeter florim porcelain slab as compared to the conventional 12-millimeter florim porcelain slab without the structural core board.

(108) It is important to note the testing results from Example 2 and Example 3 support a finding of at least 50% impact resistance increase, by using a structural core board on either a conventional 6-millimeter florim porcelain slab or a conventional 12-millimeter florim porcelain slab. Therefore, it may be understood the impact resistance is advantageously increased by at least 50% and other common values mentioned above.

Example 4

(109) In a fourth example, an adhesive bonding test was performed using three types of KANEKA MS Polymer adhesives, referred to as 1) TAW, 2) SKIF-28, and 3) SWC-008. The three adhesives were tested for tensile strength, such as a pulling test, under the International Standard Organization (ISO) standard ISO13007-2 covering ceramic tiles including grouts and adhesives, which evaluates short term results, such as cohesive failure (CF) between ceramic surfaces using grouts and adhesives, as would be understood by one skilled in the art. Each adhesive was applied to a separate mortar substrate and adhered to green foam, as would be understood by one skilled in the art. The bond was tested after 4 days. The force to separate the bond was measured for each of the adhesives is shown in the below table.

(110) TABLE-US-00004 TABLE 4 Tensile Testing Results. TAW SKIF-28 SWC-008 Green Foam Tensile Testing ISO Mortar TB MPa 0.34 0.49 0.35 ISO13007-2 Failure CF CF CF

(111) Table 4 illustrates the tensile strength of each of the adhesives. The pressure, in megapascal (MPa), for TAW was 0.34 to achieve CF. The pressure for SKIF-28 was 0.49 MPa to achieve CF. The pressure for SWC-008B was 0.35 MPa to achieve cohesive failure. SKIF-28 was observed to tolerate the highest pressure on the tensile test.

Example 5

(112) In a fifth example, a fast grab test was performed using two of the KANEKA MS PolymerTM adhesives in Example 4. The adhesives include 1) TAW and 2) SKIF-28. The two adhesives were tested for an evaluation of how much weight can be held by the substrate just after applying adhesive. One skilled in the art may refer to this test as a “non-sag” test. The force to move the

substrate was evaluated against ANSI 118.4. The results of the fast grab test are shown in the below table.

(113) TABLE-US-00005 TABLE 5 Fast Grab Test Results. TAW SKIF-28 ANSI 118.4 Fast Grab Testing 100% g/cm.sup.2 7.17 6.39 >1.94 4 inch × 5 inch coverage lb/ft.sup.2 14.68 13.10 >3.97

(114) Table 5 illustrates the results of the fast grab test in which both TAW and SKIF-28 adhesives exceed the requirements of the ANSI 118.4 standard. For example, ANSI 118.4 criteria requires more than 3.97 lb/ft.sup.2 and TAW exhibited a 14.68 lb/ft.sup.2 fast grab results. Similarly, SKIF-28 exceeded the 3.97 lb/ft.sup.2 ANSI 118.4 criteria at 13.10 lb/ft.sup.2. Porcelain tile weight may be about 2.9 lb/ft.sup.2, therefore TAW adhesive may require only about 20% coverage to hold tile or composite tile.

Example 6

(115) In a sixth example, SWC-008 adhesive, as discussed in Example 4, was tested for satisfactory bonding of the porcelain to a foam panel. The SWC-008 adhesive was tested as an offset of GLAX, as SWC-008 is a 1-part, lower viscosity, rollable, and quickdry, as compared to the mixing requirement of a 2 part system. Dry time was observed to be less than 3.5 hours. The lower viscosity of SWC-008 enables bonding to bond porcelain tile to foam panels without gaps. Adhesion strength was empirically found to be satisfactory. In a subsequent tensile test, the foam panel was deformed before the SWC-008 adhesive was peeled off. Thus, SWC-008 has been found to have sufficient and satisfactory adhesion to bond porcelain and a foam panel, as disclosed above.

Example 7

(116) In a seventh example, the compatibility of the KANEKA MS Polymer adhesives, referred to as 1) TAW, 2) SKIF-28, and 3) SWC-008, with waterproofing products, such as Redguard®, or waterstop, was tested. Each of the adhesives was applied to a wall coated with a waterproofing product. No damage or degradation of the waterproofing membrane, or wall coating, was observed.

Example 8

(117) In an eighth example, a ceiling tile test is to be performed. At the time of filing this application, a ceiling tile test is planned by first bonding porcelain and a foam panel, as discussed above, by SWC-008 and then using the composite to test durability of the adhesive in a simulated downward facing orientation to perform as a ceiling tile would be positioned, as would be understood by one skilled in the art.

(118) This application is a continuation of U.S. Non-Provisional application Ser. No. 19/013,292, filed Jan. 8, 2025, titled “REINFORCED PORCELAIN PANEL PRODUCT FABRICATION METHODS FOR ENHANCED STRUCTURAL PROTECTION,” which is a divisional of U.S. Non-Provisional application Ser. No. 18/902,121, filed Sep. 30, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT FABRICATION METHODS FOR ENHANCED STRUCTURAL PROTECTION,” now U.S. Pat. No. 12,241,258, issued Mar. 4, 2025, which is a continuation-in-part of U.S. Non-Provisional application Ser. No. 18/821,454, filed Aug. 30, 2024, titled “REINFORCED PORCELAIN PANEL SYSTEM AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” now U.S. Pat. No. 12,247,403, issued Mar. 11, 2024, which claims priority to and the benefit of U.S. Provisional Application No. 63/676,400, filed Jul. 28, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/651,803, filed May 24, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/643,778, filed May 7, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/551,903, filed Feb. 9, 2024, titled “MAGNETIC SHOWER WALL ACCESSORIES,” U.S. Provisional Application No. 63/549,820, filed Feb. 5, 2024, titled “PORCELAIN BOARD TILES,” U.S. Provisional Application No. 63/549,704, filed Feb. 5, 2024, titled “PORCELAIN BOARD FOR FABRICATORS OR

DISTRIBUTORS,” and U.S. Provisional Application No. 63/541,981, filed Oct. 2, 2023, titled “SYSTEMS AND METHODS FOR PORCELAIN BOARD,” the disclosures of all of which are incorporated herein by reference in their entireties. U.S. Non-Provisional application Ser. No. 18/902,121 is also a continuation-in-part of U.S. Non-Provisional application Ser. No. 18/821,478, filed Aug. 30, 2024, titled “REINFORCED PORCELAIN PANEL SYSTEM AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” which claims priority to and the benefit of U.S. Provisional Application No. 63/676,400, filed Jul. 28, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/651,803, filed May 24, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/643,778, filed May 7, 2024, titled “REINFORCED PORCELAIN PANEL PRODUCT AND ASSOCIATED METHODS FOR ENHANCED STRUCTURAL PROTECTION,” U.S. Provisional Application No. 63/551,903, filed Feb. 9, 2024, titled “MAGNETIC SHOWER WALL ACCESSORIES,” U.S. Provisional Application No. 63/549,820, filed Feb. 5, 2024, titled “PORCELAIN BOARD TILES,” U.S. Provisional Application No. 63/549,704, filed Feb. 5, 2024, titled “PORCELAIN BOARD FOR FABRICATORS OR DISTRIBUTORS,” and U.S. Provisional Application No. 63/541,981, filed Oct. 2, 2023, titled “SYSTEMS AND METHODS FOR PORCELAIN BOARD,” the disclosures of all of which are incorporated herein by reference in their entireties.

(119) Other objects, features, and advantages of the disclosure will become apparent from the foregoing figures, detailed description, and embodiments. It should be understood, however, that the figures, detailed description, and embodiments, while indicating specific embodiments of the disclosure, are given by way of illustration only and are not meant to be limiting. Additionally, it is contemplated that changes and modifications within the spirit and scope of the disclosure will become apparent to those skilled in the art from the detailed description. In further embodiments, features from specific embodiments may be combined with features from other embodiments. For example, features from one embodiment may be combined with features from any of the other embodiment. In further examples, additional features may be added to the specific embodiment described herein.

Claims

1. A method to fabricate one or more reinforced porcelain panel products for renovations or new constructions, the method comprising: pre-sizing a porcelain slab to have a thickness within a range between, and including, about 3-millimeters to about 8-millimeters; applying an adhesive material to a top surface of a structural core board, the structural core board including a foam-core material, a bottom surface opposite the top surface, and a total surface area defined by summing surface areas from all surfaces of the structural core board; positioning the structural core board such that the top surface of the structural core board aligns with a preselected position of a bottom surface of the porcelain slab; and abuttingly contacting the structural core board to the bottom surface of the porcelain slab with the adhesive material therebetween at the preselected position to produce a reinforced porcelain panel product, the structural core board having a thickness greater than the porcelain slab such that the total surface area of the structural core board is greater than the total surface area of the porcelain slab, the bottom surface of the structural core board (a) having a first surface plane substantially parallel to a top surface of the porcelain slab and (b) to be attached to a mounting surface when installed.

2. The method of claim 1, wherein applying the adhesive material includes using a spray, roller, bead applicators, or a combination thereof, and wherein the structural core board extends beyond outer peripheries of each porcelain slab by an overhang length evenly surrounding the outer

peripheries of each porcelain slab.

3. The method of claim 1, further comprising trimming outer peripheries of the structural core board, after the structural core board is positioned onto the porcelain slab, such that the structural core board extends an equidistant distance beyond all outer peripheries of a first surface area of the top surface of the porcelain slab, and wherein the method step of applying the adhesive material includes using a spray, roller, bead applicators, or a combination thereof, and wherein the adhesive material includes a silyl-terminated polyether base resin.

4. The method of claim 1, wherein the structural core board has a fracture toughness greater than the porcelain slab, wherein the fracture toughness of the structural core board resists a fracture after the porcelain slab is divided into two or more portions, and wherein outer peripheries of the structural core board substantially align with outer peripheries of each porcelain slab such that outer peripheries of the structural core board are substantially flush aligned with the outer peripheries of each porcelain slab, and wherein the structural core board is defined as a single layer foam-core material.

5. The method of claim 1, further comprising forming the foam-core material by one or more of the steps of molding, pressing, or layering, wherein the foam-core material includes a thermoplastic consisting of one or more of polyurethane, polycarbonate, polyphenylene oxide, polybutylene terephthalate, polyethylene terephthalate, or acrylonitrile butadiene styrene, wherein the foam-core material has a Young's modulus value lower than the porcelain slab, wherein the Young's modulus value of the foam-core material has a range between, and including, about 2.0 GPa to about 14.6 GPa, and the method further comprising pressing the structural core board onto the porcelain slab to spread the adhesive material therebetween, thereby to reduce air between the structural core board onto the porcelain slab and to affix the structural core board onto the porcelain slab when the adhesive material cures.

6. The method of claim 1, wherein the reinforced porcelain panel product has an impact resistance of at least 30% greater than porcelain having a thickness of 6 mm or higher, wherein the reinforced porcelain panel product has a breaking strength of at least 40% greater than porcelain having a thickness of at least 6 mm, wherein the structural core board has a Young's modulus value lower than the porcelain slab, wherein the Young's modulus value of the foam-core material has a range between, and including, about 2.0 GPa to about 14.6 GPa, and wherein the structural core board extends beyond outer peripheries of a first surface area of the top surface of the porcelain slab by an overhang length within a range between, and including, about 0.125 inch to about 0.750 inches evenly surrounding the outer peripheries of the first surface area of the porcelain slab.

7. The method of claim 1, further comprising scoring the top surface of the porcelain slab with one or more scorelines positioned along one or more selected locations across the top surface, thereby to guide controlled breaks to cut the porcelain slab into a selected pattern when a radial force is exerted on or near the scorelines.

8. The method of claim 7, further comprising: (a) cutting the porcelain slab along the one or more scorelines to make one or more tiles of the reinforced porcelain panel product; (b) positioning the one or more tiles of the reinforced porcelain panel product in a selected orientation; and (c) abuttingly affixing one or more tiles to a mounting surface.

9. A method to fabricate one or more reinforced porcelain panel products, the method comprising: pre-sizing a porcelain slab to have a thickness of, and including, about 3-millimeters or greater so as to define a pre-sized porcelain slab; applying an adhesive material to a top surface of a structural core board, the structural core board including a foam-core material, a bottom surface opposite the top surface; positioning the structural core board such that the top surface of the structural core board aligns with a preselected position of a bottom surface of the pre-sized porcelain slab; and contacting the top surface of the structural core board to the bottom surface of the pre-sized porcelain slab with the adhesive material so that the combination of the structural core board, the adhesive material, and the pre-sized porcelain slab produce a reinforced porcelain panel

product, the structural core board having a thickness greater than the pre-sized porcelain slab such that a total surface area of the structural core board is greater than a total surface area of the pre-sized porcelain slab and outer peripheries of the structural core board substantially align with outer peripheries of the pre-sized porcelain slab, thereby to enhance protection from cracking or chipping when the reinforced porcelain panel product is handled, the bottom surface of the structural core board having a first surface plane substantially parallel to the top surface of the pre-sized porcelain slab.

10. The method of claim 9, further comprising trimming outer peripheries of the structural core board, and wherein the method step of applying of the adhesive material includes using a spray, roller, bead applicators, or a combination thereof, wherein the foam-core material has a Young's modulus value lower than the porcelain slab, wherein the Young's modulus value of the foam-core material has a range between, and including, about 2.0 GPa to about 14.6 GPa, wherein the reinforced porcelain panel product has an impact resistance of at least 30% greater than porcelain having a thickness of 6 mm or higher, wherein the reinforced porcelain panel product has a breaking strength of at least 30% greater than porcelain having a thickness of at least 6 mm, and wherein the structural core board extends beyond outer peripheries of a first surface area of the top surface of the porcelain slab by an overhang length within a range between, and including, about 0.125 inch to about 0.750 inches evenly surrounding the outer peripheries of the first surface area of the porcelain slab.

11. The method of claim 9, wherein the foam-core material of the structural core board is formed by one or more of molding, pressing, or layering, and wherein the foam-core material includes a thermoplastic consisting of one or more of polyurethane, polycarbonate, polyphenylene oxide, polybutylene terephthalate, polyethylene terephthalate, or acrylonitrile butadiene styrene, and the method further comprising pressing the structural core board onto the porcelain slab to spread the adhesive material therebetween, thereby to reduce air between the structural core board onto the porcelain slab and to affix the structural core board onto the porcelain slab when the adhesive material cures.

12. The method of claim 11, wherein the structural core board comprises two or more interlocking backing segments so that each of the two or more interlocking backing segments interlock to another of the two or more interlocking backing segments, and wherein each of the two or more interlocking backing segments comprises a finger joint, a shiplap joint, or a dovetail joint each configured to adhesively interlock each of the two or more interlocking backing segments together.

13. A method to fabricate one or more reinforced porcelain panel products, the method comprising: positioning a structural core board such that a top surface of the structural core board aligns with a preselected position of a bottom surface of a porcelain slab thereby to substantially center the structural core board onto each porcelain slab, the structural core board comprising a foam-core material having a Young's modulus value in a range between, and including, about 2.0 GPa to about 14.6 GPa, the structural core board having a bottom surface opposite a top surface, and a total surface area defined by summing surface areas from all surfaces of the structural core board; applying an adhesive material to one or more of a top surface of the structural core board or a bottom surface of the porcelain slab; and abuttingly contacting the structural core board to the bottom surface of the porcelain slab with the adhesive material therebetween at the preselected position to produce a reinforced porcelain panel product, the structural core board having a thickness greater than the porcelain slab such that the total surface area of the structural core board is greater than the total surface area of the porcelain slab, thereby to enhance protection from cracking or chipping when the reinforced porcelain panel product is handled, the bottom surface of the structural core board having a first surface plane substantially parallel to a top surface of the porcelain slab.

14. The method of claim 13, wherein the structural core board has a fracture toughness greater than the porcelain slab, wherein the fracture toughness of the structural core board resists a fracture after

the porcelain slab is divided such that the structural core board remains intact when the porcelain slab is divided into two or more portions, and wherein outer peripheries of the structural core board substantially align with outer peripheries of the porcelain slab such that outer peripheries of the structural core board are substantially flush aligned with the outer peripheries of the porcelain slab.

15. The method of claim 14, wherein the foam-core material includes a thermoplastic consisting of one or more of polyurethane, polycarbonate, polyphenylene oxide, polybutylene terephthalate, polyethylene terephthalate, or acrylonitrile butadiene styrene, and the method further comprising pressing the structural core board onto the porcelain slab to spread the adhesive material therebetween, thereby to reduce air between the structural core board onto the porcelain slab and to affix the structural core board onto the porcelain slab when the adhesive material cures.

16. The method of claim 15, further comprising trimming outer peripheries of the structural core board, after pressing the structural core board onto the porcelain slab, wherein the method step of applying of the adhesive material includes using a spray, roller, bead applicators, or a combination thereof, and wherein the adhesive material includes a silyl-terminated polyether base resin.

17. The method of claim 14, wherein the reinforced porcelain panel product has an impact resistance of at least 30% greater than porcelain having a thickness of 6 mm or higher, wherein the reinforced porcelain panel product has a breaking strength of at least 40% greater than porcelain having a thickness of at least 6 mm, wherein the structural core board has a Young's modulus value lower than the porcelain slab, and wherein the structural core board extends beyond outer peripheries of a first surface area of the top surface of the porcelain slab by an overhang length within a range between, and including, about 0.125 inch to about 0.750 inches evenly surrounding the outer peripheries of the first surface area of the porcelain slab.

18. The method of claim 13, further comprising: (a) cutting the porcelain slab along one or more scorelines to make one or more tiles of the reinforced porcelain panel product; (b) positioning the one or more tiles of the reinforced porcelain panel product in a selected orientation; and (c) abuttingly affixing one or more tiles to a mounting surface.

19. The method of claim 13, wherein the structural core board comprises two or more interlocking backing segments so that each of the two or more interlocking backing segments interlock to another of the two or more interlocking backing segments, and wherein each of the two or more interlocking backing segments comprises a finger joint, a shiplap joint, or a dovetail joint each configured to adhesively interlock each of the two or more interlocking backing segments together.

20. The method of claim 13, wherein applying the adhesive material includes using a spray, roller, bead applicators, or a combination thereof, and wherein the adhesive material includes a silyl-terminated polyether base resin, and the method further comprising pre-sizing the porcelain slab to have a thickness within a range between, and including, about 3-millimeters to about 8-millimeters prior to applying the adhesive material to the top surface of the structural core board.
